



Threagile

Agile Threat Modeling

Threat Model Report

VaultNote Threat Model

18 April 2026

VaultNote Security Team

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About Threagile

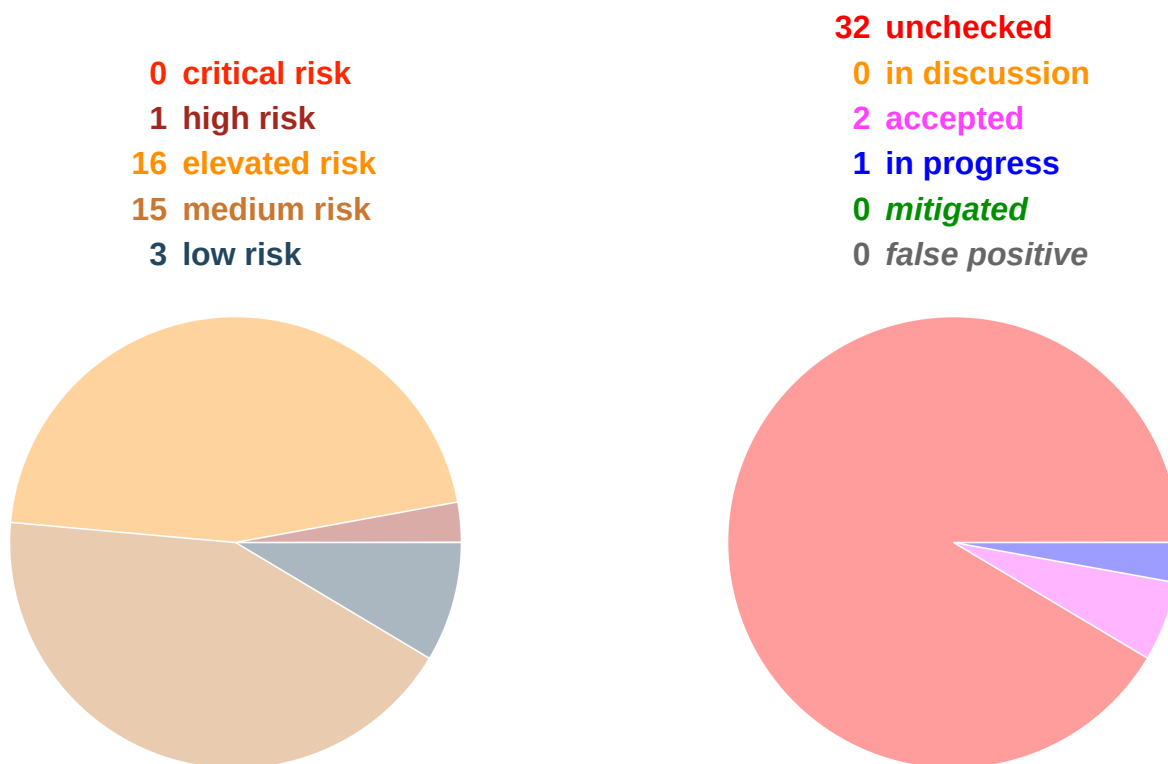
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Management Summary

Threagile toolkit was used to model the architecture of "VaultNote Threat Model" and derive risks by analyzing the components and data flows. The risks identified during this analysis are shown in the following chapters. Identified risks during threat modeling do not necessarily mean that the vulnerability associated with this risk actually exists: it is more to be seen as a list of potential risks and threats, which should be individually reviewed and reduced by removing false positives. For the remaining risks it should be checked in the design and implementation of "VaultNote Threat Model" whether the mitigation advices have been applied or not.

Each risk finding references a chapter of the OWASP ASVS (Application Security Verification Standard) audit checklist. The OWASP ASVS checklist should be considered as an inspiration by architects and developers to further harden the application in a Defense-in-Depth approach. Additionally, for each risk finding a link towards a matching OWASP Cheat Sheet or similar with technical details about how to implement a mitigation is given.

In total **35 initial risks** in **18 categories** have been identified during the threat modeling process:



VaultNote is a containerized note-taking application comprised of a React SPA, Nginx reverse proxy, Node.js/Express REST API, PostgreSQL database, Redis session store, and MinIO object storage. The architecture deliberately contains several intentional misconfigurations (unencrypted internal HTTP, no encryption at rest, hardcoded credentials) to demonstrate automated threat detection via Threagile. All findings below are expected and serve as demonstration targets.

Impact Analysis of 35 Initial Risks in 18 Categories

The most prevalent impacts of the **35 initial risks** (distributed over **18 risk categories**) are (taking the severity ratings into account and using the highest for each category):

Risk finding paragraphs are clickable and link to the corresponding chapter.

High: [SQL/NoSQL-Injection](#): 3 Initial Risks - Exploitation likelihood is *Very Likely with High impact*. If this risk is unmitigated, attackers might be able to modify SQL/NoSQL queries to steal and modify data and eventually further escalate towards a deeper system penetration via code executions.

Elevated: [Hardcoded Credentials in Container Configuration](#): 1 Initial Risk - Exploitation likelihood is *Likely with High impact*.

An attacker who obtains the compose file or inspects container environment variables gains immediate authenticated access to the affected datastore without further exploitation.

Elevated: [Missing Authentication](#): 1 Initial Risk - Exploitation likelihood is *Likely with High impact*. If this risk is unmitigated, attackers might be able to access or modify sensitive data in an unauthenticated way.

Elevated: [Missing Cloud Hardening](#): 1 Initial Risk - Exploitation likelihood is *Unlikely with Very High impact*.

If this risk is unmitigated, attackers might access cloud components in an unintended way.

Elevated: [Missing Hardening](#): 3 Initial Risks - Exploitation likelihood is *Likely with Medium impact*. If this risk remains unmitigated, attackers might be able to easier attack high-value targets.

Elevated: [Path-Traversal](#): 1 Initial Risk - Exploitation likelihood is *Very Likely with Medium impact*. If this risk is unmitigated, attackers might be able to read sensitive files (configuration data, key/credential files, deployment files, business data files, etc.) from the filesystem of affected components.

Elevated: [SQL/NoSQL-Injection](#): 3 Initial Risks - Exploitation likelihood is *Very Likely with High impact*.

If this risk is unmitigated, attackers might be able to modify SQL/NoSQL queries to steal and modify data and eventually further escalate towards a deeper system penetration via code executions.

Elevated: [Server-Side Request Forgery \(SSRF\)](#): 2 Initial Risks - Exploitation likelihood is *Likely with Medium impact*.

If this risk is unmitigated, attackers might be able to access sensitive services or files of network-reachable components by modifying outgoing calls of affected components.

Elevated: [Unencrypted Communication](#): 4 Initial Risks - Exploitation likelihood is *Likely with High impact*.

If this risk is unmitigated, network attackers might be able to eavesdrop on unencrypted sensitive data sent between components.

Elevated: Unguarded Direct Datastore Access: 3 Initial Risks - Exploitation likelihood is *Likely* with *Medium* impact.

If this risk is unmitigated, attackers might be able to directly attack sensitive data stores without any protecting components in-between.

Medium: Container Base Image Backdooring: 5 Initial Risks - Exploitation likelihood is *Unlikely* with *High* impact.

If this risk is unmitigated, attackers might be able to deeply persist in the target system by executing code in deployed containers.

Medium: Missing Build Infrastructure: 1 Initial Risk - Exploitation likelihood is *Unlikely* with *Medium* impact.

If this risk is unmitigated, attackers might be able to exploit risks unseen in this threat model due to critical build infrastructure components missing in the model.

Medium: Missing Hardening: 3 Initial Risks - Exploitation likelihood is *Likely* with *Medium* impact.

If this risk remains unmitigated, attackers might be able to easier attack high-value targets.

Medium: Missing Identity Store: 1 Initial Risk - Exploitation likelihood is *Unlikely* with *Medium* impact.

If this risk is unmitigated, attackers might be able to exploit risks unseen in this threat model in the identity provider/store that is currently missing in the model.

Medium: Missing Two-Factor Authentication (2FA): 1 Initial Risk - Exploitation likelihood is *Unlikely* with *Medium* impact.

If this risk is unmitigated, attackers might be able to access or modify highly sensitive data without strong authentication.

Medium: Missing Vault (Secret Storage): 1 Initial Risk - Exploitation likelihood is *Unlikely* with *Medium* impact.

If this risk is unmitigated, attackers might be able to easier steal config secrets (like credentials, private keys, client certificates, etc.) once a vulnerability to access files is present and exploited.

Medium: Missing Web Application Firewall (WAF): 1 Initial Risk - Exploitation likelihood is *Unlikely* with *Medium* impact.

If this risk is unmitigated, attackers might be able to apply standard attack pattern tests at great speed without any filtering.

Medium: Mixed Targets on Shared Runtime: 1 Initial Risk - Exploitation likelihood is *Unlikely* with *Medium* impact.

If this risk is unmitigated, attackers successfully attacking other components of the system might have an easy path towards more valuable targets, as they are running on the same shared runtime.

Medium: Unencrypted Technical Assets: 2 Initial Risks - Exploitation likelihood is *Unlikely* with *High* impact.

If this risk is unmitigated, attackers might be able to access unencrypted data when successfully compromising sensitive components.

Medium: Unguarded Direct Datastore Access: 3 Initial Risks - Exploitation likelihood is *Likely* with *Medium* impact.

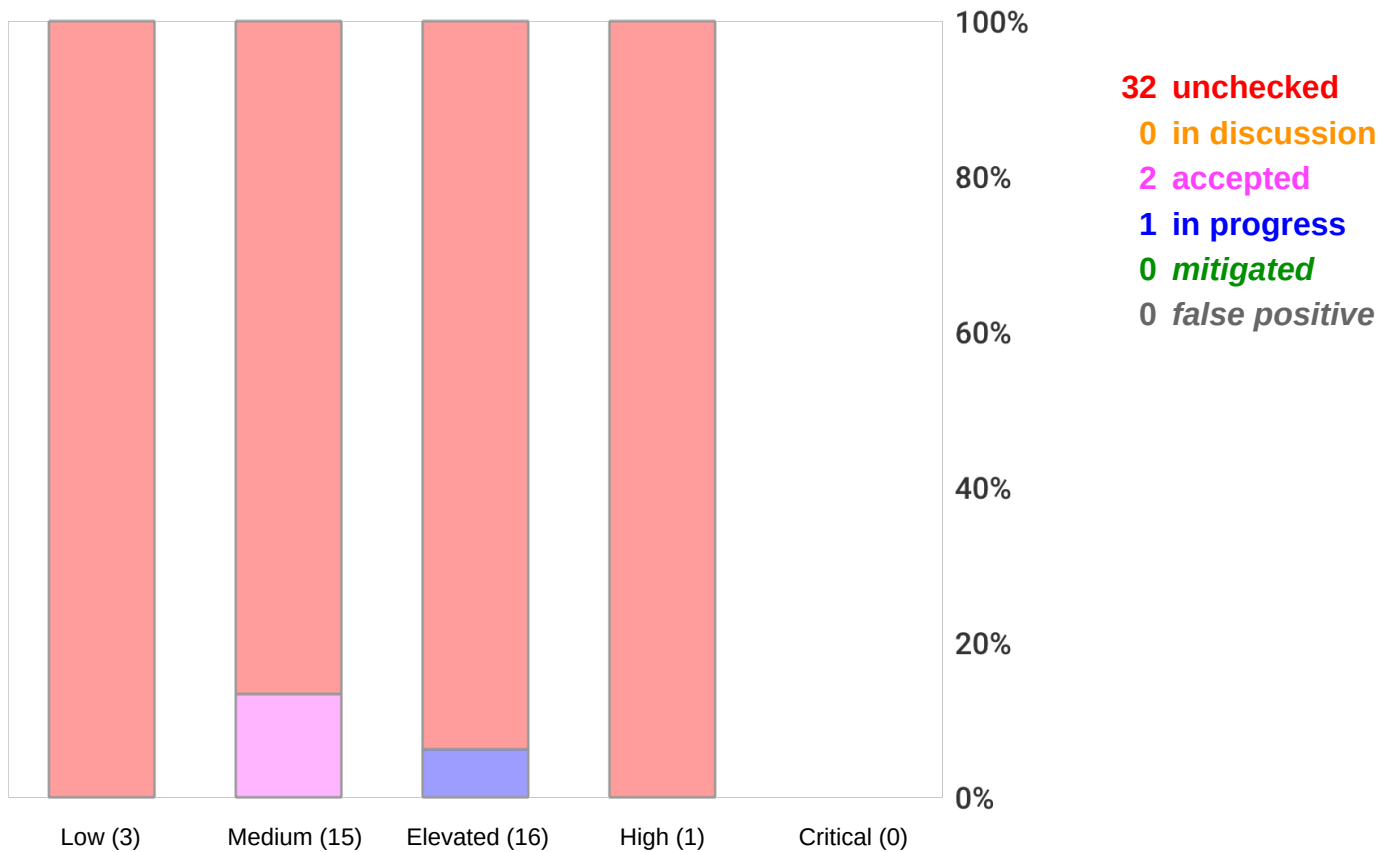
If this risk is unmitigated, attackers might be able to directly attack sensitive data stores without any protecting components in-between.

Low: DoS-risky Access Across Trust-Boundary: 3 Initial Risks - Exploitation likelihood is *Unlikely* with *Low* impact.

If this risk remains unmitigated, attackers might be able to disturb the availability of important parts of the system.

Risk Mitigation

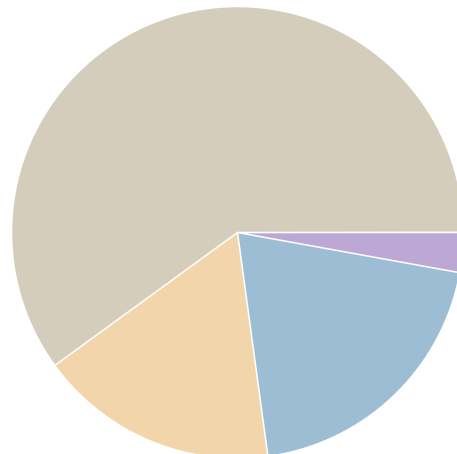
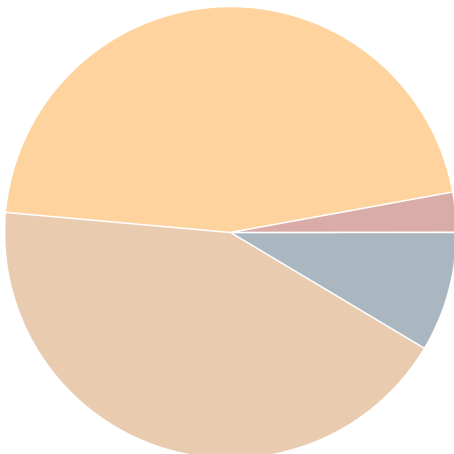
The following chart gives a high-level overview of the risk tracking status (including mitigated risks):



After removal of risks with status *mitigated* and *false positive* the following **35** remain unmitigated:

0 unmitigated critical risk
1 unmitigated high risk
16 unmitigated elevated risk
15 unmitigated medium risk
3 unmitigated low risk

1 business side related
7 architecture related
6 development related
21 operations related



Asset Register

Technical Assets

API Server

Node.js/Express REST API. Handles authentication (JWT/bcrypt), CRUD on notes, file uploads to MinIO. Bridges the frontend and backend networks. Single most security-critical component — compromise equals full data access.

Browser SPA

React Single Page Application served from Nginx. Users authenticate, manage notes, and upload files through this interface.

MinIO Object Storage

S3-compatible object store for file attachments. INTENTIONAL MISCONFIGURATION: Uses default minioadmin/minioadmin credentials (visible in docker-compose.yml) and stores files without encryption at rest. HTTP-only (no TLS) within the Docker network.

Nginx Reverse Proxy

Nginx terminates TLS and forwards all API traffic to the Express server over plain HTTP (port 3000). This is the DMZ entry point.

PostgreSQL Database

Primary relational datastore. Contains users table (email + bcrypt password hash), notes table (title + content), and attachments metadata. INTENTIONAL MISCONFIGURATION: No encryption at rest (encryption: none).

Redis Cache

In-memory store used as session store and JWT revocation list. INTENTIONAL MISCONFIGURATION: No encryption at rest. If Redis persistence is enabled and the RDB file is exfiltrated, session tokens are exposed.

Data Assets

File Attachments

Binary file uploads attached to notes. May include documents, images, or other sensitive files depending on note usage.

Note Content

User-authored note text — may include personal, financial, or health information depending on how users employ the application.

Session Tokens

JWT tokens and Redis session keys used to maintain authenticated state. Theft allows session hijacking without knowing the user's password.

User Credentials

Email addresses and bcrypt-hashed passwords used for authentication. Compromise allows full account takeover.

Impact Analysis of 35 Remaining Risks in 18 Categories

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Medium: Missing Vault (Secret Storage): 1 Remaining Risk - Exploitation likelihood is *Unlikely with Medium impact*.

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Low: **DoS-risky Access Across Trust-Boundary**: 3 Remaining Risks - Exploitation likelihood is *Unlikely* with *Low* impact.

If this risk remains unmitigated, attackers might be able to disturb the availability of important parts of the system.

Application Overview

Business Criticality

The overall business criticality of "VaultNote Threat Model" was rated as:

(archive | operational | **IMPORTANT** | critical | mission-critical)

Business Overview

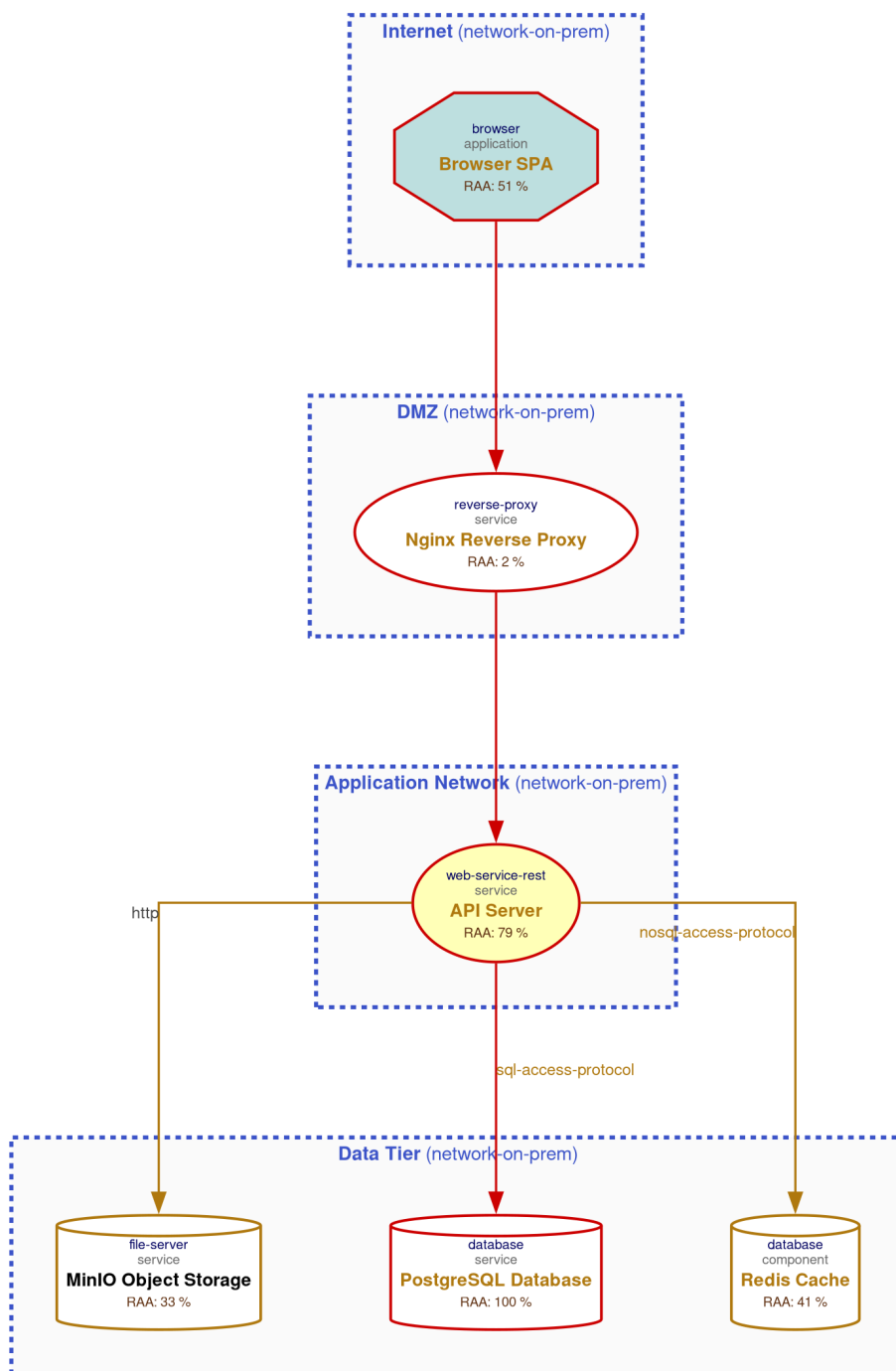
VaultNote allows authenticated users to create, edit, and share encrypted notes with file attachments. The application stores sensitive user data and note content that must be protected against unauthorized disclosure.

Technical Overview

Multi-tier containerized deployment: Nginx terminates TLS and proxies requests to the API over unencrypted HTTP. The API accesses PostgreSQL, Redis, and MinIO over plaintext TCP/HTTP within an internal Docker network.

Data-Flow Diagram

The following diagram was generated by Threagile based on the model input and gives a high-level overview of the data-flow between technical assets. The RAA value is the calculated *Relative Attacker Attractiveness* in percent. For a full high-resolution version of this diagram please refer to the PNG image file alongside this report.



Security Requirements

This chapter lists the custom security requirements which have been defined for the modeled target.

This list is not complete and regulatory or law relevant security requirements have to be taken into account as well. Also custom individual security requirements might exist for the project.

Abuse Cases

This chapter lists the custom abuse cases which have been defined for the modeled target.

This list is not complete and regulatory or law relevant abuse cases have to be taken into account as well. Also custom individual abuse cases might exist for the project.

Tag Listing

This chapter lists what tags are used by which elements.

cache

Redis Cache

container-runtime

Docker Host

docker

Docker Host

express

API Server

intentional-misconfiguration

MinIO File Storage, MinIO Object Storage, HTTP to API Server

jwt

API Server

minio

MinIO Object Storage

nginx

Nginx Reverse Proxy

nodejs

API Server

object-storage

MinIO Object Storage

postgresql

PostgreSQL Database

react

Browser SPA

redis

Redis Cache

relational

PostgreSQL Database

s3

MinIO Object Storage

session-store

Redis Cache

spa

Browser SPA

tls-termination

Nginx Reverse Proxy

STRIDE Classification of Identified Risks

This chapter clusters and classifies the risks by STRIDE categories: In total **35 potential risks** have been identified during the threat modeling process of which **1 in the Spoofing** category, **14 in the Tampering** category, **0 in the Repudiation** category, **11 in the Information Disclosure** category, **3 in the Denial of Service** category, and **6 in the Elevation of Privilege** category.

Risk finding paragraphs are clickable and link to the corresponding chapter.

Spoofing

Medium: **Missing Identity Store**: 1 / 1 Risk - Exploitation likelihood is *Unlikely with Medium impact*.

The modeled architecture does not contain an identity store, which might be the risk of a model missing critical assets (and thus not seeing their risks).

Tampering

High: **SQL/NoSQL-Injection**: 3 / 3 Risks - Exploitation likelihood is *Very Likely with High impact*.

When a database is accessed via database access protocols SQL/NoSQL-Injection risks might arise. The risk rating depends on the sensitivity technical asset itself and of the data assets processed.

Elevated: **Missing Cloud Hardening**: 1 / 1 Risk - Exploitation likelihood is *Unlikely with Very High impact*.

Cloud components should be hardened according to the cloud vendor best practices. This affects their configuration, auditing, and further areas.

Elevated: **Missing Hardening**: 3 / 3 Risks - Exploitation likelihood is *Likely with Medium impact*.

Technical assets with a Relative Attacker Attractiveness (RAA) value of 55 % or higher should be explicitly hardened taking best practices and vendor hardening guides into account.

Elevated: **SQL/NoSQL-Injection**: 3 / 3 Risks - Exploitation likelihood is *Very Likely with High impact*.

When a database is accessed via database access protocols SQL/NoSQL-Injection risks might arise. The risk rating depends on the sensitivity technical asset itself and of the data assets processed.

Medium: **Container Base Image Backdooring**: 5 / 5 Risks - Exploitation likelihood is *Unlikely with High impact*.

When a technical asset is built using container technologies, Base Image Backdooring risks might arise where base images and other layers used contain vulnerable components or backdoors.

Medium: **Missing Build Infrastructure**: 1 / 1 Risk - Exploitation likelihood is *Unlikely with Medium impact*.

The modeled architecture does not contain a build infrastructure (devops-client, sourcecode-repo, build-pipeline, etc.), which might be the risk of a model missing critical assets (and thus not

seeing their risks). If the architecture contains custom-developed parts, the pipeline where code gets developed and built needs to be part of the model.

Medium: Missing Hardening: 3 / 3 Risks - Exploitation likelihood is *Likely* with *Medium* impact.
Technical assets with a Relative Attacker Attractiveness (RAA) value of 55 % or higher should be explicitly hardened taking best practices and vendor hardening guides into account.

Medium: Missing Web Application Firewall (WAF): 1 / 1 Risk - Exploitation likelihood is *Unlikely* with *Medium* impact.

To have a first line of filtering defense, security architectures with web-services or web-applications should include a WAF in front of them. Even though a WAF is not a replacement for security (all components must be secure even without a WAF) it adds another layer of defense to the overall system by delaying some attacks and having easier attack alerting through it.

Repudiation

n/a

Information Disclosure

Elevated: Hardcoded Credentials in Container Configuration: 1 / 1 Risk - Exploitation likelihood is *Likely* with *High* impact.

Container deployments that rely on default or hardcoded credentials (e.g. minioadmin/minioadmin visible in docker-compose.yml) expose datastores to any process that can read the configuration.

Elevated: Path-Traversal: 1 / 1 Risk - Exploitation likelihood is *Very Likely* with *Medium* impact.

When a filesystem is accessed Path-Traversal or Local-File-Inclusion (LFI) risks might arise. The risk rating depends on the sensitivity of the technical asset itself and of the data assets processed.

Elevated: Server-Side Request Forgery (SSRF): 2 / 2 Risks - Exploitation likelihood is *Likely* with *Medium* impact.

When a server system (i.e. not a client) is accessing other server systems via typical web protocols Server-Side Request Forgery (SSRF) or Local-File-Inclusion (LFI) or Remote-File-Inclusion (RFI) risks might arise.

Elevated: Unencrypted Communication: 4 / 4 Risks - Exploitation likelihood is *Likely* with *High* impact.

Due to the confidentiality and/or integrity rating of the data assets transferred over the communication link this connection must be encrypted.

Medium: Missing Vault (Secret Storage): 1 / 1 Risk - Exploitation likelihood is *Unlikely* with *Medium* impact.

In order to avoid the risk of secret leakage via config files (when attacked through vulnerabilities being able to read files like Path-Traversal and others), it is best practice to use a separate hardened process with proper authentication, authorization, and audit logging to access config

secrets (like credentials, private keys, client certificates, etc.). This component is usually some kind of Vault.

Medium: **Unencrypted Technical Assets: 2 / 2 Risks** - Exploitation likelihood is *Unlikely* with *High* impact.

Due to the confidentiality rating of the technical asset itself and/or the stored data assets this technical asset must be encrypted. The risk rating depends on the sensitivity technical asset itself and of the data assets stored.

Denial of Service

Low: **DoS-risky Access Across Trust-Boundary: 3 / 3 Risks** - Exploitation likelihood is *Unlikely* with *Low* impact.

Assets accessed across trust boundaries with critical or mission-critical availability rating are more prone to Denial-of-Service (DoS) risks.

Elevation of Privilege

Elevated: **Missing Authentication: 1 / 1 Risk** - Exploitation likelihood is *Likely* with *High* impact.

Technical assets (especially multi-tenant systems) should authenticate incoming requests when the asset processes sensitive data.

Elevated: **Unguarded Direct Datastore Access: 3 / 3 Risks** - Exploitation likelihood is *Likely* with *Medium* impact.

Data stores accessed across trust boundaries must be guarded by some protecting service or application.

Medium: **Missing Two-Factor Authentication (2FA): 1 / 1 Risk** - Exploitation likelihood is *Unlikely* with *Medium* impact.

Technical assets (especially multi-tenant systems) should authenticate incoming requests with two-factor (2FA) authentication when the asset processes or stores highly sensitive data (in terms of confidentiality, integrity, and availability) and is accessed by humans.

Medium: **Mixed Targets on Shared Runtime: 1 / 1 Risk** - Exploitation likelihood is *Unlikely* with *Medium* impact.

Different attacker targets (like frontend and backend/datastore components) should not be running on the same shared (underlying) runtime.

Medium: **Unguarded Direct Datastore Access: 3 / 3 Risks** - Exploitation likelihood is *Likely* with *Medium* impact.

Data stores accessed across trust boundaries must be guarded by some protecting service or application.

Assignment by Function

This chapter clusters and assigns the risks by functions which are most likely able to check and mitigate them: In total **35 potential risks** have been identified during the threat modeling process of which **1 should be checked by Business Side**, **7 should be checked by Architecture**, **6 should be checked by Development**, and **21 should be checked by Operations**.

Risk finding paragraphs are clickable and link to the corresponding chapter.

Business Side

Medium: **Missing Two-Factor Authentication (2FA)**: 1 / 1 Risk - Exploitation likelihood is *Unlikely with Medium impact*.

Apply an authentication method to the technical asset protecting highly sensitive data via two-factor authentication for human users.

Architecture

Elevated: **Missing Authentication**: 1 / 1 Risk - Exploitation likelihood is *Likely with High impact*.

Apply an authentication method to the technical asset. To protect highly sensitive data consider the use of two-factor authentication for human users.

Elevated: **Unguarded Direct Datastore Access**: 3 / 3 Risks - Exploitation likelihood is *Likely with Medium impact*.

Encapsulate the datastore access behind a guarding service or application.

Medium: **Missing Build Infrastructure**: 1 / 1 Risk - Exploitation likelihood is *Unlikely with Medium impact*.

Include the build infrastructure in the model.

Medium: **Missing Identity Store**: 1 / 1 Risk - Exploitation likelihood is *Unlikely with Medium impact*.

Include an identity store in the model if the application has a login.

Medium: **Missing Vault (Secret Storage)**: 1 / 1 Risk - Exploitation likelihood is *Unlikely with Medium impact*.

Consider using a Vault (Secret Storage) to securely store and access config secrets (like credentials, private keys, client certificates, etc.).

Medium: **Unguarded Direct Datastore Access**: 3 / 3 Risks - Exploitation likelihood is *Likely with Medium impact*.

Encapsulate the datastore access behind a guarding service or application.

Development

High: SQL/NoSQL-Injection: 3 / 3 Risks - Exploitation likelihood is *Very Likely* with *High* impact. Try to use parameter binding to be safe from injection vulnerabilities. When a third-party product is used instead of custom developed software, check if the product applies the proper mitigation and ensure a reasonable patch-level.

Elevated: Path-Traversal: 1 / 1 Risk - Exploitation likelihood is *Very Likely* with *Medium* impact. Before accessing the file cross-check that it resides in the expected folder and is of the expected type and filename/suffix. Try to use a mapping if possible instead of directly accessing by a filename which is (partly or fully) provided by the caller. When a third-party product is used instead of custom developed software, check if the product applies the proper mitigation and ensure a reasonable patch-level.

Elevated: SQL/NoSQL-Injection: 3 / 3 Risks - Exploitation likelihood is *Very Likely* with *High* impact. Try to use parameter binding to be safe from injection vulnerabilities. When a third-party product is used instead of custom developed software, check if the product applies the proper mitigation and ensure a reasonable patch-level.

Elevated: Server-Side Request Forgery (SSRF): 2 / 2 Risks - Exploitation likelihood is *Likely* with *Medium* impact. Try to avoid constructing the outgoing target URL with caller controllable values. Alternatively use a mapping (whitelist) when accessing outgoing URLs instead of creating them including caller controllable values. When a third-party product is used instead of custom developed software, check if the product applies the proper mitigation and ensure a reasonable patch-level.

Operations

Elevated: Hardcoded Credentials in Container Configuration: 1 / 1 Risk - Exploitation likelihood is *Likely* with *High* impact. Replace hardcoded credentials with secrets managed by a vault (e.g. HashiCorp Vault, AWS Secrets Manager). Rotate default credentials before any production deployment and enforce secrets scanning in CI to prevent re-introduction.

Elevated: Missing Cloud Hardening: 1 / 1 Risk - Exploitation likelihood is *Unlikely* with *Very High* impact. Apply hardening of all cloud components and services, taking special care to follow the individual risk descriptions (which depend on the cloud provider tags in the model).

Elevated: Missing Hardening: 3 / 3 Risks - Exploitation likelihood is *Likely* with *Medium* impact. Try to apply all hardening best practices (like CIS benchmarks, OWASP recommendations, vendor recommendations, DevSec Hardening Framework, DBSAT for Oracle databases, and others).

Elevated: Unencrypted Communication: 4 / 4 Risks - Exploitation likelihood is *Likely* with *High* impact. Apply transport layer encryption to the communication link.

Medium: Container Base Image Backdooring: 5 / 5 Risks - Exploitation likelihood is *Unlikely* with *High* impact.

Apply hardening of all container infrastructures (see for example the *CIS-Benchmarks for Docker and Kubernetes* and the *Docker Bench for Security*). Use only trusted base images of the original vendors, verify digital signatures and apply image creation best practices. Also consider using Google's *Distroless* base images or otherwise very small base images. Regularly execute container image scans with tools checking the layers for vulnerable components.

Medium: Missing Hardening: 3 / 3 Risks - Exploitation likelihood is *Likely* with *Medium* impact.

Try to apply all hardening best practices (like CIS benchmarks, OWASP recommendations, vendor recommendations, DevSec Hardening Framework, DBSAT for Oracle databases, and others).

Medium: Missing Web Application Firewall (WAF): 1 / 1 Risk - Exploitation likelihood is *Unlikely* with *Medium* impact.

Consider placing a Web Application Firewall (WAF) in front of the web-services and/or web-applications. For cloud environments many cloud providers offer pre-configured WAFs. Even reverse proxies can be enhanced by a WAF component via ModSecurity plugins.

Medium: Mixed Targets on Shared Runtime: 1 / 1 Risk - Exploitation likelihood is *Unlikely* with *Medium* impact.

Use separate runtime environments for running different target components or apply similar separation styles to prevent load- or breach-related problems originating from one more attacker-facing asset impacts also the other more critical rated backend/datastore assets.

Medium: Unencrypted Technical Assets: 2 / 2 Risks - Exploitation likelihood is *Unlikely* with *High* impact.

Apply encryption to the technical asset.

Low: DoS-risky Access Across Trust-Boundary: 3 / 3 Risks - Exploitation likelihood is *Unlikely* with *Low* impact.

Apply anti-DoS techniques like throttling and/or per-client load blocking with quotas. Also for maintenance access routes consider applying a VPN instead of public reachable interfaces. Generally applying redundancy on the targeted technical asset reduces the risk of DoS.

RAA Analysis

For each technical asset the **"Relative Attacker Attractiveness"** (RAA) value was calculated in percent. The higher the RAA, the more interesting it is for an attacker to compromise the asset. The calculation algorithm takes the sensitivity ratings and quantities of stored and processed data into account as well as the communication links of the technical asset. Neighbouring assets to high-value RAA targets might receive an increase in their RAA value when they have a communication link towards that target ("Pivoting-Factor").

The following lists all technical assets sorted by their RAA value from highest (most attacker attractive) to lowest. This list can be used to prioritize on efforts relevant for the most attacker-attractive technical assets:

Technical asset paragraphs are clickable and link to the corresponding chapter.

PostgreSQL Database: RAA 100%

Primary relational datastore. Contains users table (email + bcrypt password hash), notes table (title + content), and attachments metadata. INTENTIONAL MISCONFIGURATION: No encryption at rest (encryption: none).

API Server: RAA 79%

Node.js/Express REST API. Handles authentication (JWT/bcrypt), CRUD on notes, file uploads to MinIO. Bridges the frontend and backend networks. Single most security-critical component — compromise equals full data access.

Browser SPA: RAA 51%

React Single Page Application served from Nginx. Users authenticate, manage notes, and upload files through this interface.

Redis Cache: RAA 41%

In-memory store used as session store and JWT revocation list. INTENTIONAL MISCONFIGURATION: No encryption at rest. If Redis persistence is enabled and the RDB file is exfiltrated, session tokens are exposed.

MinIO Object Storage: RAA 33%

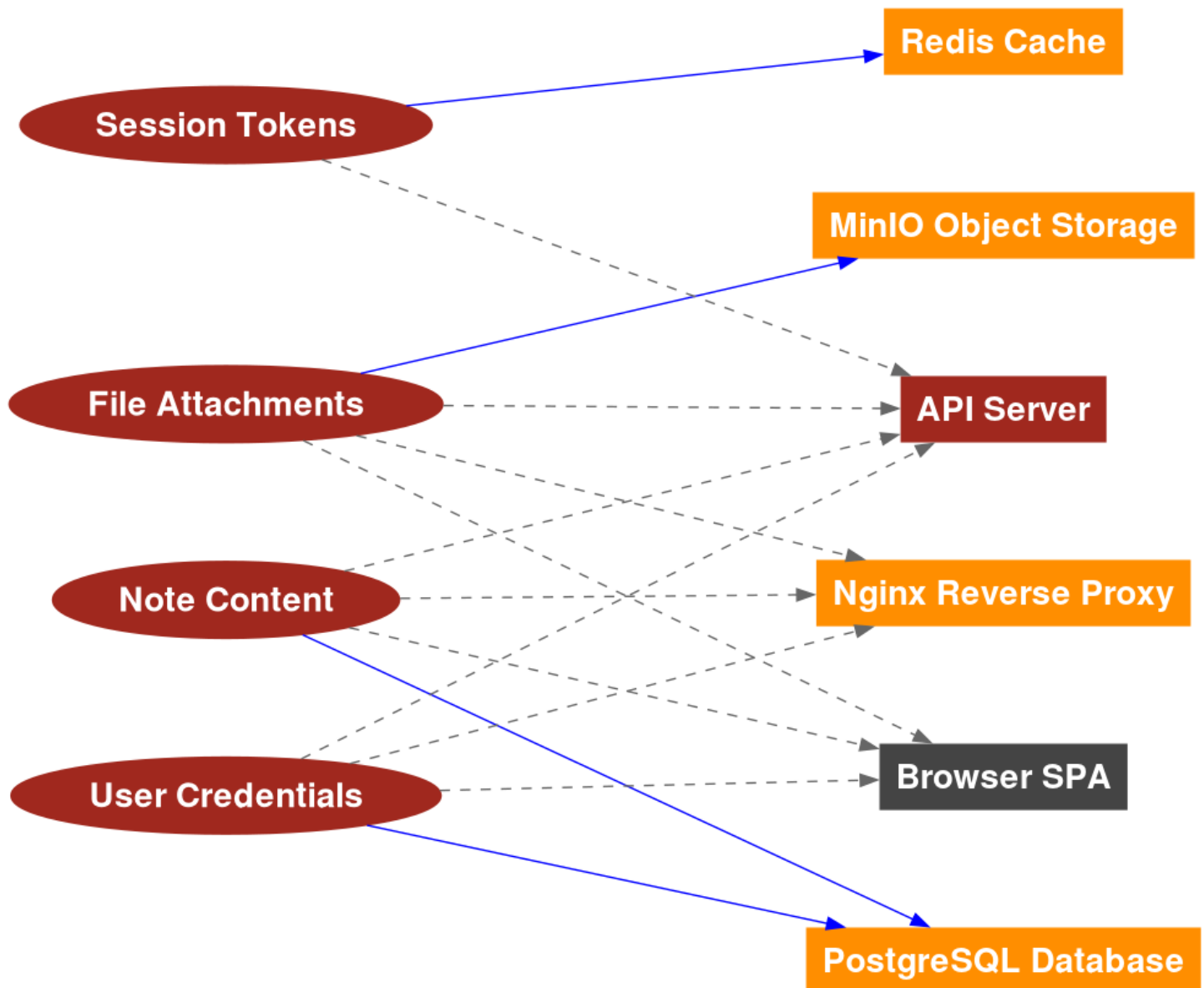
S3-compatible object store for file attachments. INTENTIONAL MISCONFIGURATION: Uses default minioadmin/minioadmin credentials (visible in docker-compose.yml) and stores files without encryption at rest. HTTP-only (no TLS) within the Docker network.

Nginx Reverse Proxy: RAA 2%

Nginx terminates TLS and forwards all API traffic to the Express server over plain HTTP (port 3000). This is the DMZ entry point.

Data Mapping

The following diagram was generated by Threagile based on the model input and gives a high-level distribution of data assets across technical assets. The color matches the identified data breach probability and risk level (see the "Data Breach Probabilities" chapter for more details). A solid line stands for *data is stored by the asset* and a dashed one means *data is processed by the asset*. For a full high-resolution version of this diagram please refer to the PNG image file alongside this report.



Out-of-Scope Assets: 0 Assets

This chapter lists all technical assets that have been defined as out-of-scope. Each one should be checked in the model whether it should better be included in the overall risk analysis:

Technical asset paragraphs are clickable and link to the corresponding chapter.

No technical assets have been defined as out-of-scope.

Potential Model Failures: 3 / 3 Risks

This chapter lists potential model failures where not all relevant assets have been modeled or the model might itself contain inconsistencies. Each potential model failure should be checked in the model against the architecture design:

Risk finding paragraphs are clickable and link to the corresponding chapter.

Medium: Missing Build Infrastructure: 1 / 1 Risk - Exploitation likelihood is *Unlikely* with *Medium* impact.

The modeled architecture does not contain a build infrastructure (devops-client, sourcecode-repo, build-pipeline, etc.), which might be the risk of a model missing critical assets (and thus not seeing their risks). If the architecture contains custom-developed parts, the pipeline where code gets developed and built needs to be part of the model.

Medium: Missing Identity Store: 1 / 1 Risk - Exploitation likelihood is *Unlikely* with *Medium* impact.

The modeled architecture does not contain an identity store, which might be the risk of a model missing critical assets (and thus not seeing their risks).

Medium: Missing Vault (Secret Storage): 1 / 1 Risk - Exploitation likelihood is *Unlikely* with *Medium* impact.

In order to avoid the risk of secret leakage via config files (when attacked through vulnerabilities being able to read files like Path-Traversal and others), it is best practice to use a separate hardened process with proper authentication, authorization, and audit logging to access config secrets (like credentials, private keys, client certificates, etc.). This component is usually some kind of Vault.

Questions: 0 / 0 Questions

This chapter lists custom questions that arose during the threat modeling process.

No custom questions arose during the threat modeling process.

Identified Risks by Vulnerability category

In total **35 potential risks** have been identified during the threat modeling process of which **0 are rated as critical, 1 as high, 16 as elevated, 15 as medium, and 3 as low.**

These risks are distributed across **18 vulnerability categories**. The following sub-chapters of this section describe each identified risk category.

SQL/NoSQL-Injection: 3 / 3 Risks

Description (Tampering): [CWE 89](#)

When a database is accessed via database access protocols SQL/NoSQL-Injection risks might arise. The risk rating depends on the sensitivity technical asset itself and of the data assets processed.

Impact

If this risk is unmitigated, attackers might be able to modify SQL/NoSQL queries to steal and modify data and eventually further escalate towards a deeper system penetration via code executions.

Detection Logic

Database accessed via typical database access protocols by in-scope clients.

Risk Rating

The risk rating depends on the sensitivity of the data stored inside the database.

False Positives

Database accesses by queries not consisting of parts controllable by the caller can be considered as false positives after individual review.

Mitigation (Development): SQL/NoSQL-Injection Prevention

Try to use parameter binding to be safe from injection vulnerabilities. When a third-party product is used instead of custom developed software, check if the product applies the proper mitigation and ensure a reasonable patch-level.

ASVS Chapter: [V5 - Validation, Sanitization and Encoding Verification Requirements](#)

Cheat Sheet: [SQL_Injection_Prevention_Cheat_Sheet](#)

Check

Are recommendations from the linked cheat sheet and referenced ASVS chapter applied?

Risk Findings

The risk **SQL/NoSQL-Injection** was found **3 times** in the analyzed architecture to be potentially possible. Each spot should be checked individually by reviewing the implementation whether all controls have been applied properly in order to mitigate each risk.

Risk finding paragraphs are clickable and link to the corresponding chapter.

High Risk Severity

SQL/NoSQL-Injection risk at **API Server** against database **PostgreSQL Database** via **PostgreSQL Connection**: Exploitation likelihood is *Very Likely* with *High* impact.

[sql-nosql-injection@api-server@postgresql-db@api-server>postgresql-connection](#)

Unchecked

Elevated Risk Severity

SQL/NoSQL-Injection risk at **API Server** against database **MinIO Object Storage** via **MinIO File Storage**: Exploitation likelihood is *Very Likely* with *Medium* impact.

[sql-nosql-injection@api-server@minio-storage@api-server>minio-file-storage](#)

Unchecked

SQL/NoSQL-Injection risk at **API Server** against database **Redis Cache** via **Redis Session Store**: Exploitation likelihood is *Very Likely* with *Medium* impact.

[sql-nosql-injection@api-server@redis-cache@api-server>redis-session-store](#)

Unchecked

Hardcoded Credentials in Container Configuration: 1 / 1 Risk

Description (Information Disclosure): [CWE 798](#)

Container deployments that rely on default or hardcoded credentials (e.g. minioadmin/minioadmin visible in docker-compose.yml) expose datastores to any process that can read the configuration.

Impact

An attacker who obtains the compose file or inspects container environment variables gains immediate authenticated access to the affected datastore without further exploitation.

Detection Logic

In-scope datastore technical assets tagged with intentional-misconfiguration.

Risk Rating

Datastores tagged intentional-misconfiguration carrying confidential or stricter data warrant at least an elevated severity because credential theft directly enables data exfiltration.

False Positives

Demo or development environments where intentional misconfigurations are documented and not internet-reachable may treat this as accepted risk.

Mitigation (Operations): Credential Rotation and Secrets Management

Replace hardcoded credentials with secrets managed by a vault (e.g. HashiCorp Vault, AWS Secrets Manager). Rotate default credentials before any production deployment and enforce secrets scanning in CI to prevent re-introduction.

ASVS Chapter: [V2 - Authentication Verification Requirements](#)

Cheat Sheet: [Secrets_Management_Cheat_Sheet](#)

Check

Are all datastore credentials sourced from a secrets manager rather than compose files or env vars?

Risk Findings

The risk **Hardcoded Credentials in Container Configuration** was found **1 time** in the analyzed architecture to be potentially possible. Each spot should be checked individually by reviewing the implementation whether all controls have been applied properly in order to mitigate each risk.

Risk finding paragraphs are clickable and link to the corresponding chapter.

Elevated Risk Severity

Hardcoded Credentials detected on **MinIO Object Storage**: credentials visible in container configuration: Exploitation likelihood is *Likely* with *High* impact.

`hardcoded-credentials@minio-storage`

Unchecked

Missing Authentication: 1 / 1 Risk

Description (Elevation of Privilege): [CWE 306](#)

Technical assets (especially multi-tenant systems) should authenticate incoming requests when the asset processes sensitive data.

Impact

If this risk is unmitigated, attackers might be able to access or modify sensitive data in an unauthenticated way.

Detection Logic

In-scope technical assets (except load-balancer, reverse-proxy, service-registry, waf, ids, and ips and in-process calls) should authenticate incoming requests when the asset processes sensitive data. This is especially the case for all multi-tenant assets (there even non-sensitive ones).

Risk Rating

The risk rating (medium or high) depends on the sensitivity of the data sent across the communication link. Monitoring callers are exempted from this risk.

False Positives

Technical assets which do not process requests regarding functionality or data linked to end-users (customers) can be considered as false positives after individual review.

Mitigation (Architecture): Authentication of Incoming Requests

Apply an authentication method to the technical asset. To protect highly sensitive data consider the use of two-factor authentication for human users.

ASVS Chapter: [V2 - Authentication Verification Requirements](#)

Cheat Sheet: [Authentication_Cheat_Sheet](#)

Check

Are recommendations from the linked cheat sheet and referenced ASVS chapter applied?

Risk Findings

The risk **Missing Authentication** was found **1 time** in the analyzed architecture to be potentially possible. Each spot should be checked individually by reviewing the implementation whether all controls have been applied properly in order to mitigate each risk.

Risk finding paragraphs are clickable and link to the corresponding chapter.

Elevated Risk Severity

Missing Authentication covering communication link **HTTP to API Server** from **Nginx Reverse Proxy to API Server**: Exploitation likelihood is *Likely* with *High* impact.

[missing-authentication@nginx-proxy>http-to-api-server@nginx-proxy@api-server](#)

Unchecked

Missing Cloud Hardening: 1 / 1 Risk

Description (Tampering): [CWE 1008](#)

Cloud components should be hardened according to the cloud vendor best practices. This affects their configuration, auditing, and further areas.

Impact

If this risk is unmitigated, attackers might access cloud components in an unintended way.

Detection Logic

In-scope cloud components (either residing in cloud trust boundaries or more specifically tagged with cloud provider types).

Risk Rating

The risk rating depends on the sensitivity of the technical asset itself and of the data assets processed.

False Positives

Cloud components not running parts of the target architecture can be considered as false positives after individual review.

Mitigation (Operations): Cloud Hardening

Apply hardening of all cloud components and services, taking special care to follow the individual risk descriptions (which depend on the cloud provider tags in the model).

For **Amazon Web Services (AWS)**: Follow the *CIS Benchmark for Amazon Web Services* (see also the automated checks of cloud audit tools like "PacBot", "CloudSploit", "CloudMapper", "ScoutSuite", or "Prowler AWS CIS Benchmark Tool").

For EC2 and other servers running Amazon Linux, follow the *CIS Benchmark for Amazon Linux* and switch to IMDSv2.

For S3 buckets follow the *Security Best Practices for Amazon S3* at

<https://docs.aws.amazon.com/AmazonS3/latest/dev/security-best-practices.html> to avoid accidental leakage.

Also take a look at some of these tools: <https://github.com/toniblyx/my-arsenal-of-aws-security-tools>

For **Microsoft Azure**: Follow the *CIS Benchmark for Microsoft Azure* (see also the automated checks of cloud audit tools like "CloudSploit" or "ScoutSuite").

For **Google Cloud Platform**: Follow the *CIS Benchmark for Google Cloud Computing Platform* (see also the automated checks of cloud audit tools like "*CloudSploit*" or "*ScoutSuite*").

For **Oracle Cloud Platform**: Follow the hardening best practices (see also the automated checks of cloud audit tools like "*CloudSploit*").

ASVS Chapter: [V1 - Architecture, Design and Threat Modeling Requirements](#)

Cheat Sheet: [Attack_Surface_Analysis_Cheat_Sheet](#)

Check

Are recommendations from the linked cheat sheet and referenced ASVS chapter applied?

Risk Findings

The risk **Missing Cloud Hardening** was found **1 time** in the analyzed architecture to be potentially possible. Each spot should be checked individually by reviewing the implementation whether all controls have been applied properly in order to mitigate each risk.

Risk finding paragraphs are clickable and link to the corresponding chapter.

Elevated Risk Severity

Missing Cloud Hardening risk at **Docker Host**: Exploitation likelihood is *Unlikely* with *Very High* impact.

[missing-cloud-hardening@docker-host](#)

Unchecked

Missing Hardening: 3 / 3 Risks

Description (Tampering): [CWE 16](#)

Technical assets with a Relative Attacker Attractiveness (RAA) value of 55 % or higher should be explicitly hardened taking best practices and vendor hardening guides into account.

Impact

If this risk remains unmitigated, attackers might be able to easier attack high-value targets.

Detection Logic

In-scope technical assets with RAA values of 55 % or higher. Generally for high-value targets like data stores, application servers, identity providers and ERP systems this limit is reduced to 40 %

Risk Rating

The risk rating depends on the sensitivity of the data processed in the technical asset.

False Positives

Usually no false positives.

Mitigation (Operations): System Hardening

Try to apply all hardening best practices (like CIS benchmarks, OWASP recommendations, vendor recommendations, DevSec Hardening Framework, DBSAT for Oracle databases, and others).

ASVS Chapter: [V14 - Configuration Verification Requirements](#)

Cheat Sheet: [Attack Surface Analysis Cheat Sheet](#)

Check

Are recommendations from the linked cheat sheet and referenced ASVS chapter applied?

Risk Findings

The risk **Missing Hardening** was found **3 times** in the analyzed architecture to be potentially possible. Each spot should be checked individually by reviewing the implementation whether all controls have been applied properly in order to mitigate each risk.

Risk finding paragraphs are clickable and link to the corresponding chapter.

Elevated Risk Severity

Missing Hardening risk at **API Server**: Exploitation likelihood is *Likely* with *Medium* impact.

[missing-hardening@api-server](#)

Unchecked

Missing Hardening risk at **PostgreSQL Database**: Exploitation likelihood is *Likely* with *Medium* impact.

[missing-hardening@postgresql-db](#)

Unchecked

Medium Risk Severity

Missing Hardening risk at **Redis Cache**: Exploitation likelihood is *Likely* with *Low* impact.

[missing-hardening@redis-cache](#)

Unchecked

Path-Traversal: 1 / 1 Risk

Description (Information Disclosure): [CWE 22](#)

When a filesystem is accessed Path-Traversal or Local-File-Inclusion (LFI) risks might arise. The risk rating depends on the sensitivity of the technical asset itself and of the data assets processed.

Impact

If this risk is unmitigated, attackers might be able to read sensitive files (configuration data, key/credential files, deployment files, business data files, etc.) from the filesystem of affected components.

Detection Logic

Filesystems accessed by in-scope callers.

Risk Rating

The risk rating depends on the sensitivity of the data stored inside the technical asset.

False Positives

File accesses by filenames not consisting of parts controllable by the caller can be considered as false positives after individual review.

Mitigation (Development): Path-Traversal Prevention

Before accessing the file cross-check that it resides in the expected folder and is of the expected type and filename/suffix. Try to use a mapping if possible instead of directly accessing by a filename which is (partly or fully) provided by the caller. When a third-party product is used instead of custom developed software, check if the product applies the proper mitigation and ensure a reasonable patch-level.

ASVS Chapter: [V12 - File and Resources Verification Requirements](#)

Cheat Sheet: [Input Validation Cheat Sheet](#)

Check

Are recommendations from the linked cheat sheet and referenced ASVS chapter applied?

Risk Findings

The risk **Path-Traversal** was found **1 time** in the analyzed architecture to be potentially possible. Each spot should be checked individually by reviewing the implementation whether all controls have been applied properly in order to mitigate each risk.

Risk finding paragraphs are clickable and link to the corresponding chapter.

Elevated Risk Severity

Path-Traversal risk at **API Server** against filesystem **MinIO Object Storage** via **MinIO File Storage**: Exploitation likelihood is *Very Likely* with *Medium* impact.

`path-traversal@api-server@minio-storage@api-server>minio-file-storage`

Unchecked

Server-Side Request Forgery (SSRF): 2 / 2 Risks

Description (Information Disclosure): [CWE 918](#)

When a server system (i.e. not a client) is accessing other server systems via typical web protocols Server-Side Request Forgery (SSRF) or Local-File-Inclusion (LFI) or Remote-File-Inclusion (RFI) risks might arise.

Impact

If this risk is unmitigated, attackers might be able to access sensitive services or files of network-reachable components by modifying outgoing calls of affected components.

Detection Logic

In-scope non-client systems accessing (using outgoing communication links) targets with either HTTP or HTTPS protocol.

Risk Rating

The risk rating (low or medium) depends on the sensitivity of the data assets receivable via web protocols from targets within the same network trust-boundary as well on the sensitivity of the data assets receivable via web protocols from the target asset itself. Also for cloud-based environments the exploitation impact is at least medium, as cloud backend services can be attacked via SSRF.

False Positives

Servers not sending outgoing web requests can be considered as false positives after review.

Mitigation (Development): SSRF Prevention

Try to avoid constructing the outgoing target URL with caller controllable values. Alternatively use a mapping (whitelist) when accessing outgoing URLs instead of creating them including caller controllable values. When a third-party product is used instead of custom developed software, check if the product applies the proper mitigation and ensure a reasonable patch-level.

ASVS Chapter: [V12 - File and Resources Verification Requirements](#)

Cheat Sheet: [Server_Side_Request_Forgery_Prevention_Cheat_Sheet](#)

Check

Are recommendations from the linked cheat sheet and referenced ASVS chapter applied?

Risk Findings

The risk **Server-Side Request Forgery (SSRF)** was found **2 times** in the analyzed architecture to be potentially possible. Each spot should be checked individually by reviewing the implementation whether all controls have been applied properly in order to mitigate each risk.

Risk finding paragraphs are clickable and link to the corresponding chapter.

Elevated Risk Severity

Server-Side Request Forgery (SSRF) risk at **API Server** server-side web-requesting the target **MinIO Object Storage** via **MinIO File Storage**: Exploitation likelihood is *Likely* with *Medium* impact.

`server-side-request-forgery@api-server@minio-storage@api-server>minio-file-storage`

Unchecked

Server-Side Request Forgery (SSRF) risk at **Nginx Reverse Proxy** server-side web-requesting the target **API Server** via **HTTP to API Server**: Exploitation likelihood is *Likely* with *Medium* impact.

`server-side-request-forgery@nginx-proxy@api-server@nginx-proxy>http-to-api-server`

Unchecked

Unencrypted Communication: 4 / 4 Risks

Description (Information Disclosure): [CWE 319](#)

Due to the confidentiality and/or integrity rating of the data assets transferred over the communication link this connection must be encrypted.

Impact

If this risk is unmitigated, network attackers might be able to eavesdrop on unencrypted sensitive data sent between components.

Detection Logic

Unencrypted technical communication links of in-scope technical assets (excluding monitoring traffic as well as local-file-access, in-process-library-call and inter-process-communication) transferring sensitive data.

Risk Rating

Depending on the confidentiality rating of the transferred data-assets either medium or high risk.

False Positives

When all sensitive data sent over the communication link is already fully encrypted on document or data level. Also intra-container/pod communication can be considered false positive when container orchestration platform handles encryption.

Mitigation (Operations): Encryption of Communication Links

Apply transport layer encryption to the communication link.

ASVS Chapter: [V9 - Communication Verification Requirements](#)

Cheat Sheet: [Transport Layer Protection Cheat Sheet](#)

Check

Are recommendations from the linked cheat sheet and referenced ASVS chapter applied?

Risk Findings

The risk **Unencrypted Communication** was found **4 times** in the analyzed architecture to be potentially possible. Each spot should be checked individually by reviewing the implementation whether all controls have been applied properly in order to mitigate each risk.

Risk finding paragraphs are clickable and link to the corresponding chapter.

Elevated Risk Severity

Unencrypted Communication named **MinIO File Storage** between **API Server** and **MinIO Object Storage** transferring authentication data (like credentials, token, session-id, etc.): Exploitation likelihood is *Likely* with *High* impact.

`unencrypted-communication@api-server>minio-file-storage@api-server@minio-storage`

Unchecked

Unencrypted Communication named **PostgreSQL Connection** between **API Server** and **PostgreSQL Database** transferring authentication data (like credentials, token, session-id, etc.): Exploitation likelihood is *Likely* with *High* impact.

`unencrypted-communication@api-server>postgresql-connection@api-server@postgresql-db`

Unchecked

Unencrypted Communication named **Redis Session Store** between **API Server** and **Redis Cache** transferring authentication data (like credentials, token, session-id, etc.): Exploitation likelihood is *Likely* with *High* impact.

`unencrypted-communication@api-server>redis-session-store@api-server@redis-cache`

Unchecked

Unencrypted Communication named **HTTP to API Server** between **Nginx Reverse Proxy** and **API Server**: Exploitation likelihood is *Likely* with *Medium* impact.

`unencrypted-communication@nginx-proxy>http-to-api-server@nginx-proxy@api-server`

In Progress

2026-04-18 Security Team

VAULT-102

Migrating Nginx-to-API communication to mTLS. A separate CA will be provisioned inside the Docker network. PR #47 is in review; target completion sprint 22.

Unguarded Direct Datastore Access: 3 / 3 Risks

Description (Elevation of Privilege): [CWE 501](#)

Data stores accessed across trust boundaries must be guarded by some protecting service or application.

Impact

If this risk is unmitigated, attackers might be able to directly attack sensitive data stores without any protecting components in-between.

Detection Logic

In-scope technical assets of type datastore (except identity-store-ldap when accessed from identity-provider and file-server when accessed via file transfer protocols) with confidentiality rating of confidential (or higher) or with integrity rating of critical (or higher) which have incoming data-flows from assets outside across a network trust-boundary. DevOps config and deployment access is excluded from this risk.

Risk Rating

The matching technical assets are at low risk. When either the confidentiality rating is strictly-confidential or the integrity rating is mission-critical, the risk-rating is considered medium. For assets with RAA values higher than 40 % the risk-rating increases.

False Positives

When the caller is considered fully trusted as if it was part of the datastore itself.

Mitigation (Architecture): Encapsulation of Datastore

Encapsulate the datastore access behind a guarding service or application.

ASVS Chapter: [V1 - Architecture, Design and Threat Modeling Requirements](#)

Cheat Sheet: [Attack_Surface_Analysis_Cheat_Sheet](#)

Check

Are recommendations from the linked cheat sheet and referenced ASVS chapter applied?

Risk Findings

The risk **Unguarded Direct Datastore Access** was found **3 times** in the analyzed architecture to be potentially possible. Each spot should be checked individually by reviewing the implementation whether all controls have been applied properly in order to mitigate each risk.

Risk finding paragraphs are clickable and link to the corresponding chapter.

Elevated Risk Severity

Unguarded Direct Datastore Access of PostgreSQL Database by API Server via PostgreSQL Connection: Exploitation likelihood is *Likely* with *Medium* impact.

[unguarded-direct-datastore-access@api-server>postgresql-connection@api-server@postgresql-db](#)

Unchecked

Unguarded Direct Datastore Access of Redis Cache by API Server via Redis Session Store: Exploitation likelihood is *Likely* with *Medium* impact.

[unguarded-direct-datastore-access@api-server>redis-session-store@api-server@redis-cache](#)

Unchecked

Medium Risk Severity

Unguarded Direct Datastore Access of MinIO Object Storage by API Server via MinIO File Storage: Exploitation likelihood is *Likely* with *Low* impact.

[unguarded-direct-datastore-access@api-server>minio-file-storage@api-server@minio-storage](#)

Unchecked

Container Base Image Backdooring: 5 / 5 Risks

Description (Tampering): [CWE 912](#)

When a technical asset is built using container technologies, Base Image Backdooring risks might arise where base images and other layers used contain vulnerable components or backdoors.

See for example:

<https://techcrunch.com/2018/06/15/tainted-crypto-mining-containers-pulled-from-docker-hub/>

Impact

If this risk is unmitigated, attackers might be able to deeply persist in the target system by executing code in deployed containers.

Detection Logic

In-scope technical assets running as containers.

Risk Rating

The risk rating depends on the sensitivity of the technical asset itself and of the data assets.

False Positives

Fully trusted (i.e. reviewed and cryptographically signed or similar) base images of containers can be considered as false positives after individual review.

Mitigation (Operations): Container Infrastructure Hardening

Apply hardening of all container infrastructures (see for example the *CIS-Benchmarks for Docker and Kubernetes* and the *Docker Bench for Security*). Use only trusted base images of the original vendors, verify digital signatures and apply image creation best practices. Also consider using Google's *Distroless* base images or otherwise very small base images. Regularly execute container image scans with tools checking the layers for vulnerable components.

ASVS Chapter: [V10 - Malicious Code Verification Requirements](#)

Cheat Sheet: [Docker_Security_Cheat_Sheet](#)

Check

Are recommendations from the linked cheat sheet and referenced ASVS/CSVS applied?

Risk Findings

The risk **Container Base Image Backdooring** was found **5 times** in the analyzed architecture to be potentially possible. Each spot should be checked individually by reviewing the implementation whether all controls have been applied properly in order to mitigate each risk.

Risk finding paragraphs are clickable and link to the corresponding chapter.

Medium Risk Severity

Container Base Image Backdooring risk at **API Server**: Exploitation likelihood is *Unlikely* with *High* impact.

[container-baseimage-backdooring@api-server](#)

Unchecked

Container Base Image Backdooring risk at **Nginx Reverse Proxy**: Exploitation likelihood is *Unlikely* with *High* impact.

[container-baseimage-backdooring@nginx-proxy](#)

Unchecked

Container Base Image Backdooring risk at **PostgreSQL Database**: Exploitation likelihood is *Unlikely* with *High* impact.

[container-baseimage-backdooring@postgresql-db](#)

Unchecked

Container Base Image Backdooring risk at **MinIO Object Storage**: Exploitation likelihood is *Unlikely* with *Medium* impact.

[container-baseimage-backdooring@minio-storage](#)

Unchecked

Container Base Image Backdooring risk at **Redis Cache**: Exploitation likelihood is *Unlikely* with *Medium* impact.

[container-baseimage-backdooring@redis-cache](#)

Unchecked

Missing Build Infrastructure: 1 / 1 Risk

Description (Tampering): [CWE 1127](#)

The modeled architecture does not contain a build infrastructure (devops-client, sourcecode-repo, build-pipeline, etc.), which might be the risk of a model missing critical assets (and thus not seeing their risks). If the architecture contains custom-developed parts, the pipeline where code gets developed and built needs to be part of the model.

Impact

If this risk is unmitigated, attackers might be able to exploit risks unseen in this threat model due to critical build infrastructure components missing in the model.

Detection Logic

Models with in-scope custom-developed parts missing in-scope development (code creation) and build infrastructure components (devops-client, sourcecode-repo, build-pipeline, etc.).

Risk Rating

The risk rating depends on the highest sensitivity of the in-scope assets running custom-developed parts.

False Positives

Models not having any custom-developed parts can be considered as false positives after individual review.

Mitigation (Architecture): Build Pipeline Hardening

Include the build infrastructure in the model.

ASVS Chapter: [V1 - Architecture, Design and Threat Modeling Requirements](#)

Cheat Sheet: [Attack_Surface_Analysis_Cheat_Sheet](#)

Check

Are recommendations from the linked cheat sheet and referenced ASVS chapter applied?

Risk Findings

The risk **Missing Build Infrastructure** was found **1 time** in the analyzed architecture to be potentially possible. Each spot should be checked individually by reviewing the implementation whether all controls have been applied properly in order to mitigate each risk.

Risk finding paragraphs are clickable and link to the corresponding chapter.

Medium Risk Severity

Missing Build Infrastructure in the threat model (referencing asset **API Server** as an example): Exploitation likelihood is *Unlikely* with *Medium* impact.

[missing-build-infrastructure@api-server](#)

Unchecked

Missing Identity Store: 1 / 1 Risk

Description (Spoofing): [CWE 287](#)

The modeled architecture does not contain an identity store, which might be the risk of a model missing critical assets (and thus not seeing their risks).

Impact

If this risk is unmitigated, attackers might be able to exploit risks unseen in this threat model in the identity provider/store that is currently missing in the model.

Detection Logic

Models with authenticated data-flows authorized via end user identity missing an in-scope identity store.

Risk Rating

The risk rating depends on the sensitivity of the end user-identity authorized technical assets and their data assets processed.

False Positives

Models only offering data/services without any real authentication need can be considered as false positives after individual review.

Mitigation (Architecture): Identity Store

Include an identity store in the model if the application has a login.

ASVS Chapter: [V2 - Authentication Verification Requirements](#)

Cheat Sheet: [Authentication_Cheat_Sheet](#)

Check

Are recommendations from the linked cheat sheet and referenced ASVS chapter applied?

Risk Findings

The risk **Missing Identity Store** was found **1 time** in the analyzed architecture to be potentially possible. Each spot should be checked individually by reviewing the implementation whether all controls have been applied properly in order to mitigate each risk.

Risk finding paragraphs are clickable and link to the corresponding chapter.

Medium Risk Severity

Missing Identity Store in the threat model (referencing asset **Nginx Reverse Proxy** as an example): Exploitation likelihood is *Unlikely* with *Medium* impact.

[missing-identity-store@nginx-proxy](#)

Unchecked

Missing Two-Factor Authentication (2FA): 1 / 1 Risk

Description (Elevation of Privilege): [CWE 308](#)

Technical assets (especially multi-tenant systems) should authenticate incoming requests with two-factor (2FA) authentication when the asset processes or stores highly sensitive data (in terms of confidentiality, integrity, and availability) and is accessed by humans.

Impact

If this risk is unmitigated, attackers might be able to access or modify highly sensitive data without strong authentication.

Detection Logic

In-scope technical assets (except load-balancer, reverse-proxy, waf, ids, and ips) should authenticate incoming requests via two-factor authentication (2FA) when the asset processes or stores highly sensitive data (in terms of confidentiality, integrity, and availability) and is accessed by a client used by a human user.

Risk Rating

medium

False Positives

Technical assets which do not process requests regarding functionality or data linked to end-users (customers) can be considered as false positives after individual review.

Mitigation (Business Side): Authentication with Second Factor (2FA)

Apply an authentication method to the technical asset protecting highly sensitive data via two-factor authentication for human users.

ASVS Chapter: [V2 - Authentication Verification Requirements](#)

Cheat Sheet: [Multifactor_Authentication_Cheat_Sheet](#)

Check

Are recommendations from the linked cheat sheet and referenced ASVS chapter applied?

Risk Findings

The risk **Missing Two-Factor Authentication (2FA)** was found **1 time** in the analyzed architecture to be potentially possible. Each spot should be checked individually by reviewing the implementation whether all controls have been applied properly in order to mitigate each risk.

Risk finding paragraphs are clickable and link to the corresponding chapter.

Medium Risk Severity

Missing Two-Factor Authentication covering communication link **HTTP to API Server** from **Browser SPA** forwarded via **Nginx Reverse Proxy** to **API Server**: Exploitation likelihood is *Unlikely* with *Medium* impact.

[missing-authentication-second-factor@nginx-proxy>http-to-api-server@nginx-proxy@api-server](#)

Unchecked

Missing Vault (Secret Storage): 1 / 1 Risk

Description (Information Disclosure): [CWE 522](#)

In order to avoid the risk of secret leakage via config files (when attacked through vulnerabilities being able to read files like Path-Traversal and others), it is best practice to use a separate hardened process with proper authentication, authorization, and audit logging to access config secrets (like credentials, private keys, client certificates, etc.). This component is usually some kind of Vault.

Impact

If this risk is unmitigated, attackers might be able to easier steal config secrets (like credentials, private keys, client certificates, etc.) once a vulnerability to access files is present and exploited.

Detection Logic

Models without a Vault (Secret Storage).

Risk Rating

The risk rating depends on the sensitivity of the technical asset itself and of the data assets processed.

False Positives

Models where no technical assets have any kind of sensitive config data to protect can be considered as false positives after individual review.

Mitigation (Architecture): Vault (Secret Storage)

Consider using a Vault (Secret Storage) to securely store and access config secrets (like credentials, private keys, client certificates, etc.).

ASVS Chapter: [V6 - Stored Cryptography Verification Requirements](#)

Cheat Sheet: [Cryptographic_Storage_Cheat_Sheet](#)

Check

GetAttribute a Vault (Secret Storage) in place?

Risk Findings

The risk **Missing Vault (Secret Storage)** was found **1 time** in the analyzed architecture to be potentially possible. Each spot should be checked individually by reviewing the implementation whether all controls have been applied properly in order to mitigate each risk.

Risk finding paragraphs are clickable and link to the corresponding chapter.

Medium Risk Severity

Missing Vault (Secret Storage) in the threat model (referencing asset **API Server** as an example): Exploitation likelihood is *Unlikely* with *Medium* impact.

[missing-vault@api-server](#)

Unchecked

Missing Web Application Firewall (WAF): 1 / 1 Risk

Description (Tampering): [CWE 1008](#)

To have a first line of filtering defense, security architectures with web-services or web-applications should include a WAF in front of them. Even though a WAF is not a replacement for security (all components must be secure even without a WAF) it adds another layer of defense to the overall system by delaying some attacks and having easier attack alerting through it.

Impact

If this risk is unmitigated, attackers might be able to apply standard attack pattern tests at great speed without any filtering.

Detection Logic

In-scope web-services and/or web-applications accessed across a network trust boundary not having a Web Application Firewall (WAF) in front of them.

Risk Rating

The risk rating depends on the sensitivity of the technical asset itself and of the data assets processed.

False Positives

Targets only accessible via WAFs or reverse proxies containing a WAF component (like ModSecurity) can be considered as false positives after individual review.

Mitigation (Operations): Web Application Firewall (WAF)

Consider placing a Web Application Firewall (WAF) in front of the web-services and/or web-applications. For cloud environments many cloud providers offer pre-configured WAFs. Even reverse proxies can be enhanced by a WAF component via ModSecurity plugins.

ASVS Chapter: [V1 - Architecture, Design and Threat Modeling Requirements](#)

Cheat Sheet: [Virtual Patching Cheat Sheet](#)

Check

GetAttribute a Web Application Firewall (WAF) in place?

Risk Findings

The risk **Missing Web Application Firewall (WAF)** was found **1 time** in the analyzed architecture to be potentially possible. Each spot should be checked individually by reviewing the implementation whether all controls have been applied properly in order to mitigate each risk.

Risk finding paragraphs are clickable and link to the corresponding chapter.

Medium Risk Severity

Missing Web Application Firewall (WAF) risk at **API Server**: Exploitation likelihood is *Unlikely* with *Medium* impact.

[missing-waf@api-server](#)

Unchecked

Mixed Targets on Shared Runtime: 1 / 1 Risk

Description (Elevation of Privilege): [CWE 1008](#)

Different attacker targets (like frontend and backend/datastore components) should not be running on the same shared (underlying) runtime.

Impact

If this risk is unmitigated, attackers successfully attacking other components of the system might have an easy path towards more valuable targets, as they are running on the same shared runtime.

Detection Logic

Shared runtime running technical assets of different trust-boundaries is at risk. Also mixing backend/datastore with frontend components on the same shared runtime is considered a risk.

Risk Rating

The risk rating (low or medium) depends on the confidentiality, integrity, and availability rating of the technical asset running on the shared runtime.

False Positives

When all assets running on the shared runtime are hardened and protected to the same extend as if all were containing/processing highly sensitive data.

Mitigation (Operations): Runtime Separation

Use separate runtime environments for running different target components or apply similar separation styles to prevent load- or breach-related problems originating from one more attacker-facing asset impacts also the other more critical rated backend/datastore assets.

ASVS Chapter: [V1 - Architecture, Design and Threat Modeling Requirements](#)

Cheat Sheet: [Attack_Surface_Analysis_Cheat_Sheet](#)

Check

Are recommendations from the linked cheat sheet and referenced ASVS chapter applied?

Risk Findings

The risk **Mixed Targets on Shared Runtime** was found **1 time** in the analyzed architecture to be potentially possible. Each spot should be checked individually by reviewing the implementation whether all controls have been applied properly in order to mitigate each risk.

Risk finding paragraphs are clickable and link to the corresponding chapter.

Medium Risk Severity

Mixed Targets on Shared Runtime named **Docker Host** might enable attackers moving from one less valuable target to a more valuable one: Exploitation likelihood is *Unlikely* with *Medium* impact.

[mixed-targets-on-shared-runtime@docker-host](#)

Accepted

2026-04-10

Infrastructure Team

VAULT-95

Docker containers run without privileged mode and with no host path mounts except the named volumes. The Docker socket is not mounted into any container. Risk is accepted for this demo environment.

Unencrypted Technical Assets: 2 / 2 Risks

Description (Information Disclosure): [CWE 311](#)

Due to the confidentiality rating of the technical asset itself and/or the stored data assets this technical asset must be encrypted. The risk rating depends on the sensitivity technical asset itself and of the data assets stored.

Impact

If this risk is unmitigated, attackers might be able to access unencrypted data when successfully compromising sensitive components.

Detection Logic

In-scope unencrypted technical assets (excluding reverse-proxy, load-balancer, waf, ids, ips and embedded components like library) storing data assets rated at least as confidential or critical. For technical assets storing data assets rated as strictly-confidential or mission-critical the encryption must be of type data-with-end-user-individual-key.

Risk Rating

Depending on the confidentiality rating of the stored data-assets either medium or high risk.

False Positives

When all sensitive data stored within the asset is already fully encrypted on document or data level.

Mitigation (Operations): Encryption of Technical Asset

Apply encryption to the technical asset.

ASVS Chapter: [V6 - Stored Cryptography Verification Requirements](#)

Cheat Sheet: [Cryptographic Storage Cheat Sheet](#)

Check

Are recommendations from the linked cheat sheet and referenced ASVS chapter applied?

Risk Findings

The risk **Unencrypted Technical Assets** was found **2 times** in the analyzed architecture to be potentially possible. Each spot should be checked individually by reviewing the implementation whether all controls have been applied properly in order to mitigate each risk.

Risk finding paragraphs are clickable and link to the corresponding chapter.

Medium Risk Severity

Unencrypted Technical Asset named **Redis Cache**: Exploitation likelihood is *Unlikely* with *Medium* impact.

[unencrypted-asset@redis-cache](#)

Unchecked

Unencrypted Technical Asset named **PostgreSQL Database** missing end user individual encryption with data-with-end-user-individual-key: Exploitation likelihood is *Unlikely* with *High* impact.

[unencrypted-asset@postgresql-db](#)

Accepted

2026-04-15 Infrastructure Team VAULT-89

The PostgreSQL volume runs on a host-level LUKS-encrypted partition in all production deployments. The Threagile finding is acknowledged as informational for this demo environment which uses unencrypted storage. Production database TDE via pgcrypto will be evaluated in Q3.

DoS-risky Access Across Trust-Boundary: 3 / 3 Risks

Description (Denial of Service): [CWE 400](#)

Assets accessed across trust boundaries with critical or mission-critical availability rating are more prone to Denial-of-Service (DoS) risks.

Impact

If this risk remains unmitigated, attackers might be able to disturb the availability of important parts of the system.

Detection Logic

In-scope technical assets (excluding load-balancer) with availability rating of critical or higher which have incoming data-flows across a network trust-boundary (excluding devops usage).

Risk Rating

Matching technical assets with availability rating of critical or higher are at low risk. When the availability rating is mission-critical and neither a VPN nor IP filter for the incoming data-flow nor redundancy for the asset is applied, the risk-rating is considered medium.

False Positives

When the accessed target operations are not time- or resource-consuming.

Mitigation (Operations): Anti-DoS Measures

Apply anti-DoS techniques like throttling and/or per-client load blocking with quotas. Also for maintenance access routes consider applying a VPN instead of public reachable interfaces. Generally applying redundancy on the targeted technical asset reduces the risk of DoS.

ASVS Chapter: [V1 - Architecture, Design and Threat Modeling Requirements](#)

Cheat Sheet: [Denial_of_Service_Cheat_Sheet](#)

Check

Are recommendations from the linked cheat sheet and referenced ASVS chapter applied?

Risk Findings

The risk **DoS-risky Access Across Trust-Boundary** was found **3 times** in the analyzed architecture to be potentially possible. Each spot should be checked individually by reviewing the implementation whether all controls have been applied properly in order to mitigate each risk.

Risk finding paragraphs are clickable and link to the corresponding chapter.

Low Risk Severity

Denial-of-Service risky access of **API Server** by **Browser SPA** via **HTTPS** to **Nginx** forwarded via **Nginx Reverse Proxy**: Exploitation likelihood is *Unlikely* with *Low* impact.

[dos-risky-access-across-trust-boundary@api-server@browser-spa@browser-spa>https-to-nginx->nginx-proxy>http-to-api-server](#)

Unchecked

Denial-of-Service risky access of **Nginx Reverse Proxy** by **Browser SPA** via **HTTPS** to **Nginx**: Exploitation likelihood is *Unlikely* with *Low* impact.

[dos-risky-access-across-trust-boundary@nginx-proxy@browser-spa@browser-spa>https-to-nginx->](#)

Unchecked

Denial-of-Service risky access of **PostgreSQL Database** by **API Server** via **PostgreSQL Connection**: Exploitation likelihood is *Unlikely* with *Low* impact.

[dos-risky-access-across-trust-boundary@postgresql-db@api-server@api-server>postgresql-connection->](#)

Unchecked

Identified Risks by Technical Asset

In total **35 potential risks** have been identified during the threat modeling process of which **0 are rated as critical, 1 as high, 16 as elevated, 15 as medium, and 3 as low.**

These risks are distributed across **6 in-scope technical assets**. The following sub-chapters of this section describe each identified risk grouped by technical asset. The RAA value of a technical asset is the calculated "Relative Attacker Attractiveness" value in percent.

API Server: 16 / 16 Risks

Description

Node.js/Express REST API. Handles authentication (JWT/bcrypt), CRUD on notes, file uploads to MinIO. Bridges the frontend and backend networks. Single most security-critical component — compromise equals full data access.

Identified Risks of Asset

Risk finding paragraphs are clickable and link to the corresponding chapter.

High Risk Severity

SQL/NoSQL-Injection risk at **API Server** against database **PostgreSQL Database** via **PostgreSQL Connection**: Exploitation likelihood is *Very Likely* with *High* impact.

`sql-nosql-injection@api-server>postgresql-db@api-server>postgresql-connection`

Unchecked

Elevated Risk Severity

Missing Authentication covering communication link **HTTP to API Server** from **Nginx Reverse Proxy to API Server**: Exploitation likelihood is *Likely* with *High* impact.

`missing-authentication@nginx-proxy>http-to-api-server@nginx-proxy@api-server`

Unchecked

Unencrypted Communication named **MinIO File Storage** between **API Server** and **MinIO Object Storage** transferring authentication data (like credentials, token, session-id, etc.): Exploitation likelihood is *Likely* with *High* impact.

`unencrypted-communication@api-server>minio-file-storage@api-server@minio-storage`

Unchecked

Unencrypted Communication named **PostgreSQL Connection** between **API Server** and **PostgreSQL Database** transferring authentication data (like credentials, token, session-id, etc.): Exploitation likelihood is *Likely* with *High* impact.

`unencrypted-communication@api-server>postgresql-connection@api-server@postgresql-db`

Unchecked

Unencrypted Communication named **Redis Session Store** between **API Server** and **Redis Cache** transferring authentication data (like credentials, token, session-id, etc.): Exploitation likelihood is *Likely* with *High* impact.

`unencrypted-communication@api-server>redis-session-store@api-server@redis-cache`

Unchecked

Path-Traversal risk at **API Server** against filesystem **MinIO Object Storage** via **MinIO File Storage**: Exploitation likelihood is *Very Likely* with *Medium* impact.

path-traversal@api-server@minio-storage@api-server>minio-file-storage

Unchecked

SQL/NoSQL-Injection risk at **API Server** against database **MinIO Object Storage** via **MinIO File Storage**: Exploitation likelihood is *Very Likely* with *Medium* impact.

sql-nosql-injection@api-server@minio-storage@api-server>minio-file-storage

Unchecked

SQL/NoSQL-Injection risk at **API Server** against database **Redis Cache** via **Redis Session Store**: Exploitation likelihood is *Very Likely* with *Medium* impact.

sql-nosql-injection@api-server@redis-cache@api-server>redis-session-store

Unchecked

Missing Hardening risk at **API Server**: Exploitation likelihood is *Likely* with *Medium* impact.

missing-hardening@api-server

Unchecked

Server-Side Request Forgery (SSRF) risk at **API Server** server-side web-requesting the target **MinIO Object Storage** via **MinIO File Storage**: Exploitation likelihood is *Likely* with *Medium* impact.

server-side-request-forgery@api-server@minio-storage@api-server>minio-file-storage

Unchecked

Medium Risk Severity

Container Base Image Backdooring risk at **API Server**: Exploitation likelihood is *Unlikely* with *High* impact.

container-baseimage-backdooring@api-server

Unchecked

Missing Build Infrastructure in the threat model (referencing asset **API Server** as an example): Exploitation likelihood is *Unlikely* with *Medium* impact.

missing-build-infrastructure@api-server

Unchecked

Missing Two-Factor Authentication covering communication link **HTTP to API Server** from **Browser SPA** forwarded via **Nginx Reverse Proxy** to **API Server**: Exploitation likelihood is *Unlikely* with *Medium* impact.

missing-authentication-second-factor@nginx-proxy>http-to-api-server@nginx-proxy@api-server

Unchecked

Missing Vault (Secret Storage) in the threat model (referencing asset **API Server** as an example): Exploitation likelihood is *Unlikely* with *Medium* impact.

missing-vault@api-server

Unchecked

Missing Web Application Firewall (WAF) risk at API Server: Exploitation likelihood is *Unlikely* with *Medium* impact.

missing-waf@api-server

Unchecked

Low Risk Severity

Denial-of-Service risky access of **API Server** by **Browser SPA** via **HTTPS** to **Nginx** forwarded via **Nginx Reverse Proxy**: Exploitation likelihood is *Unlikely* with *Low* impact.

dos-risky-access-across-trust-boundary@api-server@browser-spa@browser-spa>https-to-nginx->nginx-proxy>http-to-api-server

Unchecked

Asset Information

ID:	api-server
Type:	process
Usage:	business
RAA:	79 %
Size:	service
Technology:	web-service-rest
Tags:	express, jwt, nodejs
Internet:	false
Machine:	container
Encryption:	none
Multi-Tenant:	false
Redundant:	false
Custom-Developed:	true
Client by Human:	false
Data Processed:	File Attachments, Note Content, Session Tokens, User Credentials
Data Stored:	none
Formats Accepted:	JSON

Asset Rating

Owner:	VaultNote Team
Confidentiality:	strictly-confidential (rated 5 in scale of 5)
Integrity:	critical (rated 4 in scale of 5)
Availability:	critical (rated 4 in scale of 5)
CIA-Justification:	The API processes all user data. It bridges both Docker networks — a compromised API container can pivot into the internal data tier.

Outgoing Communication Links: 3

Target technical asset names are clickable and link to the corresponding chapter.

Redis Session Store (outgoing)

Session token storage and JWT revocation list. Password-protected (requirepass) but no TLS.

Target:	Redis Cache
Protocol:	nosql-access-protocol
Encrypted:	false
Authentication:	credentials
Authorization:	technical-user
Read-Only:	false
Usage:	business
Tags:	none
VPN:	false
IP-Filtered:	false
Data Sent:	Session Tokens
Data Received:	Session Tokens

PostgreSQL Connection (outgoing)

Application-level SQL queries for user auth, notes CRUD. Uses parameterized queries (pg library) to prevent SQL injection. No TLS on the connection — traffic is plaintext within backend-net.

Target:	PostgreSQL Database
Protocol:	sql-access-protocol
Encrypted:	false
Authentication:	credentials
Authorization:	technical-user
Read-Only:	false
Usage:	business
Tags:	none
VPN:	false
IP-Filtered:	false
Data Sent:	Note Content, User Credentials
Data Received:	Note Content, User Credentials

MinIO File Storage (outgoing)

INTENTIONAL MISCONFIGURATION: S3 API calls to MinIO over plain HTTP. Default minioadmin/minioadmin credentials are hardcoded in compose. Any container on backend-net

can read all stored files.

Target:	MinIO Object Storage
Protocol:	http
Encrypted:	false
Authentication:	credentials
Authorization:	technical-user
Read-Only:	false
Usage:	business
Tags:	intentional-misconfiguration
VPN:	false
IP-Filtered:	false
Data Sent:	File Attachments
Data Received:	File Attachments

Incoming Communication Links: 1

Source technical asset names are clickable and link to the corresponding chapter.

HTTP to API Server (incoming)

INTENTIONAL MISCONFIGURATION: Traffic between Nginx and the API server uses plaintext HTTP. An attacker with access to the Docker network (container escape, compromised sidecar) can intercept credentials, session tokens, and note content in transit.

Source:	Nginx Reverse Proxy
Protocol:	http
Encrypted:	false
Authentication:	none
Authorization:	none
Read-Only:	false
Usage:	business
Tags:	intentional-misconfiguration
VPN:	false
IP-Filtered:	false
Data Received:	File Attachments, Note Content, User Credentials
Data Sent:	File Attachments, Note Content

MinIO Object Storage: 3 / 3 Risks

Description

S3-compatible object store for file attachments. INTENTIONAL MISCONFIGURATION: Uses default minioadmin/minioadmin credentials (visible in docker-compose.yml) and stores files without encryption at rest. HTTP-only (no TLS) within the Docker network.

Identified Risks of Asset

Risk finding paragraphs are clickable and link to the corresponding chapter.

Elevated Risk Severity

Hardcoded Credentials detected on **MinIO Object Storage**: credentials visible in container configuration: Exploitation likelihood is *Likely* with *High* impact.

[hardcoded-credentials@minio-storage](#)

Unchecked

Medium Risk Severity

Container Base Image Backdooring risk at **MinIO Object Storage**: Exploitation likelihood is *Unlikely* with *Medium* impact.

[container-baseimage-backdooring@minio-storage](#)

Unchecked

Unguarded Direct Datastore Access of MinIO Object Storage by API Server via MinIO File Storage: Exploitation likelihood is *Likely* with *Low* impact.

[unguarded-direct-datastore-access@api-server>minio-file-storage@api-server@minio-storage](#)

Unchecked

Asset Information

ID:	minio-storage
Type:	datastore
Usage:	business
RAA:	33 %
Size:	service
Technology:	file-server
Tags:	intentional-misconfiguration, minio, object-storage, s3
Internet:	false
Machine:	container
Encryption:	none

Multi-Tenant:	false
Redundant:	false
Custom-Developed:	false
Client by Human:	false
Data Processed:	File Attachments
Data Stored:	File Attachments
Formats Accepted:	File

Asset Rating

Owner:	VaultNote Team	
Confidentiality:	confidential	(rated 4 in scale of 5)
Integrity:	important	(rated 3 in scale of 5)
Availability:	operational	(rated 2 in scale of 5)
CIA-Justification:	Stores user file uploads which may contain sensitive documents. Default credentials + no encryption = trivial exfiltration by any process on backend-net.	

Incoming Communication Links: 1

Source technical asset names are clickable and link to the corresponding chapter.

MinIO File Storage (incoming)

INTENTIONAL MISCONFIGURATION: S3 API calls to MinIO over plain HTTP. Default minioadmin/minioadmin credentials are hardcoded in compose. Any container on backend-net can read all stored files.

Source:	API Server
Protocol:	http
Encrypted:	false
Authentication:	credentials
Authorization:	technical-user
Read-Only:	false
Usage:	business
Tags:	intentional-misconfiguration
VPN:	false
IP-Filtered:	false
Data Received:	File Attachments
Data Sent:	File Attachments

Nginx Reverse Proxy: 5 / 5 Risks

Description

Nginx terminates TLS and forwards all API traffic to the Express server over plain HTTP (port 3000). This is the DMZ entry point.

Identified Risks of Asset

Risk finding paragraphs are clickable and link to the corresponding chapter.

Elevated Risk Severity

Server-Side Request Forgery (SSRF) risk at **Nginx Reverse Proxy** server-side web-requesting the target **API Server** via **HTTP to API Server**: Exploitation likelihood is *Likely* with *Medium* impact.

`server-side-request-forgery@nginx-proxy@api-server@nginx-proxy>http-to-api-server`

Unchecked

Unencrypted Communication named **HTTP to API Server** between **Nginx Reverse Proxy** and **API Server**: Exploitation likelihood is *Likely* with *Medium* impact.

`unencrypted-communication@nginx-proxy>http-to-api-server@nginx-proxy@api-server`

In Progress

2026-04-18

Security Team

VAULT-102

Migrating Nginx-to-API communication to mTLS. A separate CA will be provisioned inside the Docker network. PR #47 is in review; target completion sprint 22.

Medium Risk Severity

Container Base Image Backdooring risk at **Nginx Reverse Proxy**: Exploitation likelihood is *Unlikely* with *High* impact.

`container-baseimage-backdooring@nginx-proxy`

Unchecked

Missing Identity Store in the threat model (referencing asset **Nginx Reverse Proxy** as an example): Exploitation likelihood is *Unlikely* with *Medium* impact.

`missing-identity-store@nginx-proxy`

Unchecked

Low Risk Severity

Denial-of-Service risky access of **Nginx Reverse Proxy** by **Browser SPA** via **HTTPS to Nginx**: Exploitation likelihood is *Unlikely* with *Low* impact.

`dos-risky-access-across-trust-boundary@nginx-proxy@browser-spa@browser-spa>https-to-nginx->`

Unchecked

Asset Information

ID:	nginx-proxy
Type:	process
Usage:	business
RAA:	2 %
Size:	service
Technology:	reverse-proxy
Tags:	nginx, tls-termination
Internet:	false
Machine:	container
Encryption:	none
Multi-Tenant:	false
Redundant:	false
Custom-Developed:	false
Client by Human:	false
Data Processed:	File Attachments, Note Content, User Credentials
Data Stored:	none
Formats Accepted:	JSON

Asset Rating

Owner:	VaultNote Team
Confidentiality:	strictly-confidential (rated 5 in scale of 5)
Integrity:	critical (rated 4 in scale of 5)
Availability:	critical (rated 4 in scale of 5)
CIA-Justification:	Nginx is the single external entry point. Its compromise exposes all downstream services. Availability is critical — outage = full service outage.

Outgoing Communication Links: 1

Target technical asset names are clickable and link to the corresponding chapter.

HTTP to API Server (outgoing)

INTENTIONAL MISCONFIGURATION: Traffic between Nginx and the API server uses plaintext HTTP. An attacker with access to the Docker network (container escape, compromised sidecar) can intercept credentials, session tokens, and note content in transit.

Target:	API Server
---------	------------

Protocol:	http
Encrypted:	false
Authentication:	none
Authorization:	none
Read-Only:	false
Usage:	business
Tags:	intentional-misconfiguration
VPN:	false
IP-Filtered:	false
Data Sent:	File Attachments, Note Content, User Credentials
Data Received:	File Attachments, Note Content

Incoming Communication Links: 1

Source technical asset names are clickable and link to the corresponding chapter.

HTTPS to Nginx (incoming)

User-initiated HTTPS requests carrying credentials (login) and note content (create/edit). This is the only encrypted link in the chain.

Source:	Browser SPA
Protocol:	https
Encrypted:	true
Authentication:	credentials
Authorization:	end-user-identity-propagation
Read-Only:	false
Usage:	business
Tags:	none
VPN:	false
IP-Filtered:	false
Data Received:	File Attachments, Note Content, User Credentials
Data Sent:	File Attachments, Note Content

PostgreSQL Database: 5 / 5 Risks

Description

Primary relational datastore. Contains users table (email + bcrypt password hash), notes table (title + content), and attachments metadata. INTENTIONAL MISCONFIGURATION: No encryption at rest (encryption: none).

Identified Risks of Asset

Risk finding paragraphs are clickable and link to the corresponding chapter.

Elevated Risk Severity

Missing Hardening risk at **PostgreSQL Database**: Exploitation likelihood is *Likely* with *Medium* impact.

[missing-hardening@postgresql-db](#)

Unchecked

Unguarded Direct Datastore Access of PostgreSQL Database by API Server via PostgreSQL Connection: Exploitation likelihood is *Likely* with *Medium* impact.

[unguarded-direct-datastore-access@api-server>postgresql-connection@api-server@postgresql-db](#)

Unchecked

Medium Risk Severity

Container Base Image Backdooring risk at **PostgreSQL Database**: Exploitation likelihood is *Unlikely* with *High* impact.

[container-baseimage-backdooring@postgresql-db](#)

Unchecked

Unencrypted Technical Asset named **PostgreSQL Database** missing end user individual encryption with data-with-end-user-individual-key: Exploitation likelihood is *Unlikely* with *High* impact.

[unencrypted-asset@postgresql-db](#)

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The PostgreSQL volume runs on a host-level LUKS-encrypted partition in all production deployments. The Threagile finding is acknowledged as informational for this demo environment which uses unencrypted storage. Production database TDE via pgcrypto will be evaluated in Q3.

Low Risk Severity

Denial-of-Service risky access of **PostgreSQL Database** by **API Server** via **PostgreSQL Connection**: Exploitation likelihood is *Unlikely* with *Low* impact.

[dos-risky-access-across-trust-boundary@postgresql-db@api-server@api-server>postgresql-connection->](#)

Unchecked

Asset Information

ID:	postgresql-db
Type:	datastore
Usage:	business
RAA:	100 %
Size:	service
Technology:	database
Tags:	postgresql, relational
Internet:	false
Machine:	container
Encryption:	none
Multi-Tenant:	false
Redundant:	false
Custom-Developed:	false
Client by Human:	false
Data Processed:	Note Content, User Credentials
Data Stored:	Note Content, User Credentials
Formats Accepted:	JSON

Asset Rating

Owner:	VaultNote Team
Confidentiality:	strictly-confidential (rated 5 in scale of 5)
Integrity:	critical (rated 4 in scale of 5)
Availability:	critical (rated 4 in scale of 5)
CIA-Justification:	Holds all user PII (email, hashed passwords) and all note content. Strictly confidential — unencrypted disk access exposes all data.

Incoming Communication Links: 1

Source technical asset names are clickable and link to the corresponding chapter.

PostgreSQL Connection (incoming)

Application-level SQL queries for user auth, notes CRUD. Uses parameterized queries (pg library) to prevent SQL injection. No TLS on the connection — traffic is plaintext within backend-net.

Source:	API Server
---------	------------

Protocol:	sql-access-protocol
Encrypted:	false
Authentication:	credentials
Authorization:	technical-user
Read-Only:	false
Usage:	business
Tags:	none
VPN:	false
IP-Filtered:	false
Data Received:	Note Content, User Credentials
Data Sent:	Note Content, User Credentials

Redis Cache: 4 / 4 Risks

Description

In-memory store used as session store and JWT revocation list. INTENTIONAL MISCONFIGURATION: No encryption at rest. If Redis persistence is enabled and the RDB file is exfiltrated, session tokens are exposed.

Identified Risks of Asset

Risk finding paragraphs are clickable and link to the corresponding chapter.

Elevated Risk Severity

Unguarded Direct Datastore Access of Redis Cache by API Server via Redis Session Store: Exploitation likelihood is *Likely* with *Medium* impact.

[unguarded-direct-datastore-access@api-server>redis-session-store@api-server@redis-cache](#)

Unchecked

Medium Risk Severity

Container Base Image Backdooring risk at **Redis Cache**: Exploitation likelihood is *Unlikely* with *Medium* impact.

[container-baseimage-backdooring@redis-cache](#)

Unchecked

Unencrypted Technical Asset named **Redis Cache**: Exploitation likelihood is *Unlikely* with *Medium* impact.

[unencrypted-asset@redis-cache](#)

Unchecked

Missing Hardening risk at **Redis Cache**: Exploitation likelihood is *Likely* with *Low* impact.

[missing-hardening@redis-cache](#)

Unchecked

Asset Information

ID:	redis-cache
Type:	datastore
Usage:	business
RAA:	41 %
Size:	component
Technology:	database
Tags:	cache, redis, session-store

Internet:	false
Machine:	container
Encryption:	none
Multi-Tenant:	false
Redundant:	false
Custom-Developed:	false
Client by Human:	false
Data Processed:	Session Tokens
Data Stored:	Session Tokens
Formats Accepted:	none of the special data formats accepted

Asset Rating

Owner:	VaultNote Team	
Confidentiality:	confidential	(rated 4 in scale of 5)
Integrity:	critical	(rated 4 in scale of 5)
Availability:	important	(rated 3 in scale of 5)
CIA-Justification:	Session token exposure enables account takeover. Non-persistent by default but the container volume could persist RDB snapshots.	

Incoming Communication Links: 1

Source technical asset names are clickable and link to the corresponding chapter.

Redis Session Store (incoming)

Session token storage and JWT revocation list. Password-protected (requirepass) but no TLS.

Source:	API Server
Protocol:	nosql-access-protocol
Encrypted:	false
Authentication:	credentials
Authorization:	technical-user
Read-Only:	false
Usage:	business
Tags:	none
VPN:	false
IP-Filtered:	false
Data Received:	Session Tokens
Data Sent:	Session Tokens

Browser SPA: 0 / 0 Risks

Description

React Single Page Application served from Nginx. Users authenticate, manage notes, and upload files through this interface.

Identified Risks of Asset

No risksStr were identified.

Asset Information

ID:	browser-spa
Type:	external-entity
Usage:	business
RAA:	51 %
Size:	application
Technology:	browser
Tags:	react, spa
Internet:	true
Machine:	physical
Encryption:	none
Multi-Tenant:	false
Redundant:	false
Custom-Developed:	true
Client by Human:	true
Data Processed:	File Attachments, Note Content, User Credentials
Data Stored:	none
Formats Accepted:	JSON

Asset Rating

Owner:	End Users
Confidentiality:	strictly-confidential (rated 5 in scale of 5)
Integrity:	critical (rated 4 in scale of 5)
Availability:	operational (rated 2 in scale of 5)
CIA-Justification:	Client-side code is public; the trust boundary here is user data entered into the browser. XSS would allow an attacker to exfiltrate session tokens.

Outgoing Communication Links: 1

Target technical asset names are clickable and link to the corresponding chapter.

HTTPS to Nginx (outgoing)

User-initiated HTTPS requests carrying credentials (login) and note content (create/edit). This is the only encrypted link in the chain.

Target:	Nginx Reverse Proxy
Protocol:	https
Encrypted:	true
Authentication:	credentials
Authorization:	end-user-identity-propagation
Read-Only:	false
Usage:	business
Tags:	none
VPN:	false
IP-Filtered:	false
Data Sent:	File Attachments, Note Content, User Credentials
Data Received:	File Attachments, Note Content

Identified Data Breach Probabilities by Data Asset

In total **35 potential risks** have been identified during the threat modeling process of which **0 are rated as critical, 1 as high, 16 as elevated, 15 as medium, and 3 as low.**

These risks are distributed across **4 data assets**. The following sub-chapters of this section describe the derived data breach probabilities grouped by data asset.

Technical asset names and risk IDs are clickable and link to the corresponding chapter.

File Attachments: 17 / 17 Risks

Binary file uploads attached to notes. May include documents, images, or other sensitive files depending on note usage.

ID:	file-attachments
Usage:	business
Quantity:	many
Tags:	none
Origin:	customer
Owner:	VaultNote Team
Confidentiality:	confidential (rated 4 in scale of 5)
Integrity:	important (rated 3 in scale of 5)
Availability:	operational (rated 2 in scale of 5)
CIA-Justification:	Files may contain confidential content. MinIO's default credentials (minioadmin) make all stored files accessible to anyone on the network.
Processed by:	API Server, Browser SPA, MinIO Object Storage, Nginx Reverse Proxy
Stored by:	MinIO Object Storage
Sent via:	MinIO File Storage, HTTPS to Nginx, HTTP to API Server
Received via:	MinIO File Storage, HTTPS to Nginx, HTTP to API Server
Data Breach:	probable
Data Breach Risks:	This data asset has data breach potential because of 17 remaining risksStr:
	Probable: container-baseimage-backdooring@api-server
	Probable: container-baseimage-backdooring@minio-storage
	Probable: container-baseimage-backdooring@nginx-proxy
	Probable: hardcoded-credentials@minio-storage
	Probable: missing-cloud-hardening@docker-host
	Probable: path-traversal@api-server@minio-storage@api-server>minio-file-storage
	Probable: sql-nosql-injection@api-server@minio-storage@api-server>minio-file-storage
	Possible: missing-authentication@nginx-proxy>http-to-api-server@nginx-proxy@api-server
	Possible: missing-authentication-second-factor@nginx-proxy>http-to-api-server@nginx-proxy@api-server
	Possible: server-side-request-forgery@api-server@minio-storage@api-server>minio-file-storage
	Possible: server-side-request-forgery@nginx-proxy@api-server@nginx-proxy>http-to-api-server
	Possible: unencrypted-communication@api-server>minio-file-storage@api-server@minio-storage
	Possible: unencrypted-communication@nginx-proxy>http-to-api-server@nginx-proxy@api-server
	Improbable: missing-hardening@api-server
	Improbable: missing-waf@api-server
	Improbable: unguarded-direct-datastore-access@api-server>minio-file-storage@api-server@minio-storage
	Improbable: mixed-targets-on-shared-runtime@docker-host

Note Content: 17 / 17 Risks

User-authored note text — may include personal, financial, or health information depending on how users employ the application.

ID:	note-content
Usage:	business
Quantity:	many
Tags:	none
Origin:	customer
Owner:	VaultNote Team
Confidentiality:	confidential (rated 4 in scale of 5)
Integrity:	critical (rated 4 in scale of 5)
Availability:	operational (rated 2 in scale of 5)
CIA-Justification:	Notes may contain highly sensitive personal information. Integrity is critical because corrupted notes lose user trust.
Processed by:	API Server, Browser SPA, Nginx Reverse Proxy, PostgreSQL Database
Stored by:	PostgreSQL Database
Sent via:	PostgreSQL Connection, HTTPS to Nginx, HTTP to API Server
Received via:	PostgreSQL Connection, HTTPS to Nginx, HTTP to API Server
Data Breach:	probable
Data Breach Risks:	This data asset has data breach potential because of 17 remaining risksStr: - Probable: container-baseimage-backdooring@api-server - Probable: container-baseimage-backdooring@nginx-proxy - Probable: container-baseimage-backdooring@postgresql-db - Probable: missing-cloud-hardening@docker-host - Probable: sql-nosql-injection@api-server@postgresql-db@api-server>postgresql-connection - Possible: missing-authentication@nginx-proxy>http-to-api-server@nginx-proxy@api-server - Possible: missing-authentication-second-factor@nginx-proxy>http-to-api-server@nginx-proxy@api-server - Possible: server-side-request-forgery@api-server@minio-storage@api-server>minio-file-storage - Possible: server-side-request-forgery@nginx-proxy@api-server@nginx-proxy>http-to-api-server - Possible: unencrypted-communication@api-server>postgresql-connection@api-server@postgresql-db - Possible: unencrypted-communication@nginx-proxy>http-to-api-server@nginx-proxy@api-server - Improbable: missing-hardening@api-server - Improbable: missing-hardening@postgresql-db - Improbable: missing-waf@api-server - Improbable: unguarded-direct-datastore-access@api-server>postgresql-connection@api-server@postgresql-db - Improbable: mixed-targets-on-shared-runtime@docker-host - Improbable: unencrypted-asset@postgresql-db

Session Tokens: 15 / 15 Risks

JWT tokens and Redis session keys used to maintain authenticated state. Theft allows session hijacking without knowing the user's password.

ID:	session-tokens
Usage:	business
Quantity:	many
Tags:	none
Origin:	internal
Owner:	VaultNote Team
Confidentiality:	confidential (rated 4 in scale of 5)
Integrity:	critical (rated 4 in scale of 5)
Availability:	operational (rated 2 in scale of 5)
CIA-Justification:	Session token compromise enables full session hijacking. Short TTLs and revocation lists mitigate but do not eliminate the risk.
Processed by:	API Server, Redis Cache
Stored by:	Redis Cache
Sent via:	Redis Session Store
Received via:	Redis Session Store
Data Breach:	probable

Data Breach Risks: This data asset has data breach potential because of 15 remaining risksStr:

Probable: container-baseimage-backdooring@api-server
 Probable: container-baseimage-backdooring@redis-cache
 Probable: missing-cloud-hardening@docker-host
 Probable: sql-nosql-injection@api-server@redis-cache@api-server>redis-session-store
 Possible: missing-authentication@nginx-proxy>http-to-api-server@nginx-proxy@api-server
 Possible: missing-authentication-second-factor@nginx-proxy>http-to-api-server@nginx-proxy@api-server
 Possible: server-side-request-forgery@api-server@minio-storage@api-server>minio-file-storage
 Possible: unencrypted-communication@api-server>redis-session-store@api-server@redis-cache
 Possible: unencrypted-communication@nginx-proxy>http-to-api-server@nginx-proxy@api-server
 Improbable: missing-hardening@api-server
 Improbable: missing-hardening@redis-cache
 Improbable: missing-waf@api-server
 Improbable: unencrypted-asset@redis-cache
 Improbable: unguarded-direct-datastore-access@api-server>redis-session-store@api-server@redis-cache
 Improbable: mixed-targets-on-shared-runtime@docker-host

User Credentials: 17 / 17 Risks

Email addresses and bcrypt-hashed passwords used for authentication. Compromise allows full account takeover.

ID:	user-credentials
Usage:	business
Quantity:	many
Tags:	none
Origin:	customer
Owner:	VaultNote Team
Confidentiality:	strictly-confidential (rated 5 in scale of 5)
Integrity:	critical (rated 4 in scale of 5)
Availability:	operational (rated 2 in scale of 5)
CIA-Justification:	Credentials grant full account access. A breach leaks all users' identities and enables account takeover across every note and attachment.
Processed by:	API Server, Browser SPA, Nginx Reverse Proxy, PostgreSQL Database
Stored by:	PostgreSQL Database
Sent via:	PostgreSQL Connection, HTTPS to Nginx, HTTP to API Server
Received via:	PostgreSQL Connection
Data Breach:	probable
Data Breach Risks:	This data asset has data breach potential because of 17 remaining risksStr:

Probable: container-baseimage-backdooring@api-server

Probable: container-baseimage-backdooring@nginx-proxy

Probable: container-baseimage-backdooring@postgresql-db

Probable: missing-cloud-hardening@docker-host

Probable: sql-nosql-injection@api-server@postgresql-db@api-server>postgresql-connection

Possible: missing-authentication@nginx-proxy>http-to-api-server@nginx-proxy@api-server

Possible: missing-authentication-second-factor@nginx-proxy>http-to-api-server@nginx-proxy@api-server

Possible: server-side-request-forgery@api-server@minio-storage@api-server>minio-file-storage

Possible: server-side-request-forgery@nginx-proxy@api-server@nginx-proxy>http-to-api-server

Possible: unencrypted-communication@api-server>postgresql-connection@api-server@postgresql-db

Possible: unencrypted-communication@nginx-proxy>http-to-api-server@nginx-proxy@api-server

Improbable: missing-hardening@api-server

Improbable: missing-hardening@postgresql-db

Improbable: missing-waf@api-server

Improbable: unguarded-direct-datastore-access@api-server>postgresql-connection@api-server@postgresql-db

Improbable: mixed-targets-on-shared-runtime@docker-host

Improbable: unencrypted-asset@postgresql-db

Trust Boundaries

In total **4 trust boundaries** have been modeled during the threat modeling process.

Application Network

Internal Docker bridge network (frontend-net). Contains the API server which is reachable from Nginx but also bridges into backend-net.

ID: application-network
Type: network-on-prem
Tags: none
Assets inside: API Server
Boundaries nested: none

DMZ

Demilitarized zone containing the Nginx reverse proxy. Accessible from the internet but isolated from the application and data tiers.

ID: dmz
Type: network-on-prem
Tags: none
Assets inside: Nginx Reverse Proxy
Boundaries nested: none

Data Tier

Internal Docker bridge network (backend-net, marked internal: true). No external routing. Contains all datastores. The API server has a NIC on this network and acts as the sole legitimate access path.

ID: data-tier
Type: network-on-prem
Tags: none
Assets inside: MinIO Object Storage, PostgreSQL Database, Redis Cache
Boundaries nested: none

Internet

Untrusted external network. All traffic from this zone must cross the Nginx TLS termination point before entering the DMZ.

ID: internet
Type: [network-on-prem](#)
Tags: none
Assets inside: Browser SPA
Boundaries nested: none

Shared Runtimes

In total **1 shared runtime** has been modeled during the threat modeling process.

Docker Host

All containers run on the same Docker host. A container escape or privileged container could access the host filesystem including Docker socket and database volumes.

ID:	docker-host
Tags:	container-runtime, docker
Assets running:	Nginx Reverse Proxy, API Server, PostgreSQL Database, Redis Cache, MinIO Object Storage

Risk Rules Checked by Threagile

Threagile Version: 1.0.0

Threagile Build Timestamp: 20260523112448

Threagile Execution Timestamp: 20260523112503

Model Filename: /app/work/threagile.yaml

Model Hash (SHA256): 774c898d4f3b0ed271aba4c7150c9cea775c2aea5a63edcdafe444d6aa0ccd57

Threagile (see <https://threagile.io> for more details) is an open-source toolkit for agile threat modeling, created by Christian Schneider (<https://christian-schneider.net>): It allows to model an architecture with its assets in an agile fashion as a YAML file directly inside the IDE. Upon execution of the Threagile toolkit all standard risk rules (as well as individual custom rules if present) are checked against the architecture model. At the time the Threagile toolkit was executed on the model input file the following risk rules were checked:

Disclaimer

VaultNote Security Team conducted this threat analysis using the open-source Threagile toolkit on the applications and systems that were modeled as of this report's date. Information security threats are continually changing, with new vulnerabilities discovered on a daily basis, and no application can ever be 100% secure no matter how much threat modeling is conducted. It is recommended to execute threat modeling and also penetration testing on a regular basis (for example yearly) to ensure a high ongoing level of security and constantly check for new attack vectors.

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In this particular project, a time box approach was used to define the analysis effort. This means that the author allotted a prearranged amount of time to identify and document threats. Because of this, there is no guarantee that all possible threats and risks are discovered. Furthermore, the analysis applies to a snapshot of the current state of the modeled architecture (based on the architecture information provided by the customer) at the examination time.

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